

Soft and On-Shell Counterterm Scheme Goal

Phase 1 · Local soft counterterms
with on-shell counterterms deferred

Complete implementation plan, derivations, investigation record,
authoritative corrections, and final acceptance reconciliation

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Phase 1 — Local Soft Counterterms, with On-Shell Counterterms Deferred

Summary

Implement the local soft-refined counterterm operator of [arXiv:2203.11038v1](#), especially §§4.2–4.4 and Appendix B, in both GammaLoop representations:

$$H_d = U_d + S_{d-1} - U_d S_{d-1}, \quad S_{-1} = 0.$$

The implementation will:

- Cleanly defer all on-shell counterterm work behind explicit `unimplemented!` panics. No rest-frame substitution, γ^0 , or OS projectors will remain as WIP behavior.
- Complete local soft counterterms in the 4D recursive forest and the 3D CFF/residue representation.
- Apply every parent operator to the complete nested child atom, preserving the exact forest cross terms when parent and child use different expansion points.
- Keep integrated soft and OS counterterms disabled until the local implementation and nesting tests are complete.
- Add the paper's Appendix-B topology, a fully derived toy nested example, inspect-level structural checks, actual profile ultra-violet CLI tests, and a physical $q_g \rightarrow 0$ test using a massive-top bubble in a two-loop $\gamma^* \rightarrow d\bar{d}$ vertex.

1. Mandatory implementation startup

Before implementation, perform read-only preflight checks: reread `CONTRIBUTING.typ`, inspect the current worktree, and verify that no conflicting active goal exists.

The first repository mutation must write this entire `<proposed_plan>...</proposed_plan>` block verbatim, including its tags, to:

`/common/dev/gammaploop/lcnbr/SOFT_OS_CT_SCHEME_GOAL.typ`

The literal next tool action, with no intervening inspection or implementation action, must call `create_goal` with the objective:

Implement `/common/dev/gammaploop/lcnbr/SOFT_OS_CT_SCHEME_GOAL.typ`: defer local OS counterterms and complete and verify the local soft-counterterm scheme.

No implementation work starts before that goal exists.

2. Scope, dispatch, and failure policy

Connected components and schemes

- Leave `CTIdentifier` and `ApproximationType` unchanged. They already select the approximation policy at the correct level: one connected 1PI component.
- Classify every connected component first. A disconnected spinney is retained only when all of its connected components are selected; it has no independent approximation scheme.
- Apply the selected operator to each connected component and multiply completed results for disjoint components.
- Use `HedgePoset` as the reference path. Compare with the legacy path only on topologies it already supports.
- Keep the existing inert `softct` setting inert; it receives no dispatch or precedence semantics.
- Allow hybrid woods such as a soft-refined child and ordinary-MUV parent.
- Reject an IR/soft component if `generate_integrated=true`.
- Reject an IR/soft and `PolePart` mixture in the same wood until an explicit integrated policy exists.

OS cleanup

In both `local_4d.rs` and `local_3d.rs`:

- Route `ApproximationType::OS` directly to:

```
unimplemented!(
  "local on-shell counterterms are deferred until local counterterms can be derived from the 4D
  expanded representation"
)
```

- Remove incomplete OS transformations, rest-frame graph rewriting, γ^0 construction, and $P_{\pm}$ projector logic.

- Retain the enum and configuration syntax for compatibility.
- Move useful WIP comments into the goal document or the adjacent deferred-OS explanation rather than discarding them.
- Replace malformed OS graph fixtures with one minimal valid component-dispatch test which reaches the intended panic.
- Do not retain a massless-OS shortcut to the soft operator: OS is wholly deferred in this phase.

Color closure

For bare or amplitude-level fixtures with open fundamental color indices, multiply by the color-singlet projector

$$\frac{\delta_i^j}{N_c}.$$

Existing Lorentz projector machinery may close Lorentz indices automatically; color closure must be explicit. Tests must fail if an unsaturated color index remains.

3. Operator definitions and implementation contract

For a connected component δ of superficial UV degree d , define the ordinary UV deformation

$$\Phi_\delta^U(u)X = X(\{up\}, \{um\}, \{k\}; D_{e,u}),$$

with

$$D_{e,u} = k_e^2 - M^2 + 2u k_e \cdot p_e + u^2 p_e^2 - u^2 (m_e^2 - M^2).$$

Define the soft deformation

$$\Phi_\delta^S(s)X = X(\{sp\}, \{m\}, \{k\}).$$

Thus:

- U scales component-external four-momenta and physical masses and introduces the UV mass M through the existing UV propagator expansion.
- S scales only component-external four-momenta.
- S leaves loop momenta and every physical mass unchanged.
- S never introduces M .
- A terminal denominator already containing M is not reinterpreted as a physical-mass denominator by an outer operation.
- A free physical mass in a numerator remains graded by an outer U , while a literal M remains fixed.

With

$$J_z^{\leq n} f = \sum_{j=0}^n \frac{1}{j!} \partial_z^j f|_{z=0}, \quad J_z^{\leq -1} = 0,$$

the operators are

$$U_d = \text{ev}_{u=1} J_u^{\leq d} \Phi^U, \quad S_{d-1} = \text{ev}_{s=1} J_s^{\leq d-1} \Phi^S,$$

and

$$H_d = U_d + S_{d-1} - U_d S_{d-1}.$$

$U_d S_{d-1}$ means: construct and combine the complete soft Taylor polynomial first, then apply a fresh ordinary UV operation to that completed expression.

For $d = 0$,

$$H_0 = U_0.$$

4D construction

Refactor only as much as needed to reuse the existing compatible-LMB selection, raw Taylor projection, and final tensor/gamma normalization:

1. Construct raw $U_d X$.
2. For $d > 0$, construct raw $S_{d-1} X$.
3. Apply ordinary U_d to the completed soft result, producing $U_d S_{d-1} X$.
4. Form $U_d X + S_{d-1} X - U_d S_{d-1} X$.
5. Only then apply the forest subtraction sign, CT marker, and canonical simplification.

There must be one completed local atom per component branch, not separately marked UV, soft, and overlap counterterms.

3D CFF construction

Apply the same algebra to the full component-local CFF expression:

- Determine surface ownership from the connected component's canonical route and topology shore.
- Rewrite only E/H-surface occurrences owned wholly by the active component.
- Leave co-graph and mixed-boundary surfaces unchanged.
- Scale both external spatial dependence and external-energy shifts.
- Keep loop energies, loop spatial momenta, physical masses, and existing M -denominators fixed in the soft branch.
- Perform the replacement before Symbolica atom collapse whenever structural equality could otherwise merge distinct topology occurrences.
- Preserve structural equality of identical CFF denominators; do not manufacture different mathematical cache IDs merely to encode ownership.
- Carry an occurrence-provenance side table keyed by residue/forest history when equal denominators on different shores need different ownership decisions. This plan explicitly authorizes that minimal internal metadata if the existing cache provenance cannot be extended.
- Error with component, residue, surface, and route context if ownership is ambiguous. Never fall back to global replacement.

For $d > 0$, retain the fixed Laurent endpoint -1 . In the dressed CFF expression, Taylor degree j appears at Laurent power $j - d$, so $j = 0, \dots, d - 1$ is exactly the strictly negative part required for S_{d-1} . Bypass the soft branch for $d = 0$. Apply U_d to the completed CFF soft branch, combine $U + S - US$, and return one combined atom per residue. Preserve the export identity

$$(\text{forest index, node index, node key, term index, residue index})$$

through normalization.

4. Why differently centred nested jets define a valid R-operation

The implementation must be guided by these algebraic invariants rather than by assuming that all Taylor operators share a centre.

First,

$$1 - H_d = (1 - U_d)(1 - S_{d-1})$$

holds exactly, without assuming that U and S commute. Therefore:

- S removes the unwanted low-order soft behavior.
- U subsequently removes the UV behavior of the soft-completed branch.
- The difference from ordinary UV subtraction is

$$H_d - U_d = (1 - U_d)S_{d-1},$$

which is UV-improved by construction.

Do not assume that H is idempotent or that $US = SU$.

For a nested child $\gamma \subset \Gamma$, let K_γ and K_Γ denote whichever local operator, U or H , is selected for each connected component. Define

$$P_\Gamma = (1 - K_\gamma)I, \quad Z_\Gamma = -K_\Gamma P_\Gamma.$$

The nested forest result is

$$R = (1 - K_\Gamma)(1 - K_\gamma)I = I - K_\gamma I - K_\Gamma I + K_\Gamma K_\gamma I.$$

The pure outer terms combine as

$$-K_\Gamma I + K_\Gamma K_\gamma I = -K_\Gamma(1 - K_\gamma)I,$$

so the outer approximation acts on the child-subtracted integrand, not on the original integrand independently. Conversely,

$$-K_\gamma I + K_\Gamma K_\gamma I = -(1 - K_\Gamma)K_\gamma I.$$

The implementation and tests must establish

$$A_\gamma[K_\Gamma(1 - K_\gamma)I] = 0$$

for the child hard limit, and

$$A_{\Gamma/\gamma}[(1 - K_\Gamma)K_\gamma I] = 0$$

for the reduced outer hard limit.

A sufficient routing condition is

$$A_\gamma K_\Gamma = K_\Gamma A_\gamma.$$

Enforce this by using a single canonical component route based on homogeneous, unimodular loop-basis transformations. Do not use a path-dependent “first compatible LMB”, and reject an affine shift by an external momentum when it would alter which variable an outer operator regards as fixed.

The practical nested rule is:

- Finish and combine the child operator first.
- Treat the child boundary momentum as an internal loop variable when it belongs to the parent loop system.
- A parent S scales only the parent’s external momentum.
- A parent U scales the parent’s external momentum and physical masses.
- Both parent operators act on the actual completed child expression, including its polynomial dependence on the child boundary momentum.
- Existing terminal M -denominators remain terminal; physical masses still participate in an outer U .

This is mathematically sufficient for nested subdivergence cancellation under the paper’s renormalizable, nonexceptional-momentum hypotheses. Genuine overlapping, non-nested components remain separate compatible forests in HedgePoset; incompatible components are never multiplied in one forest.

The 3D implementation must additionally verify, rather than merely assume, residue compatibility:

$$\sum_{\rho} \text{Res}_{\rho}(B_{4d}I) = B_{3d} \left(\sum_{\rho} \text{Res}_{\rho}I \right), \quad B \in \{U, S, US, H\},$$

including nested compositions. Per-residue equality is required only after matching the canonical forest/residue history.

5. Explicit nested toy derivation

This derivation must be copied into the goal file and converted into term-by-term tests. Work in a $1 + 1$ -dimensional kinematic slice,

$$\ell^2 = \ell_0^2 - \ell_1^2,$$

while retaining the original four-dimensional superficial degrees for the Taylor orders.

Define

$$V_{\ell} = \ell^2 - m^2, \quad U_{\ell} = \ell^2 - M^2, \quad \Delta = M^2 - m^2,$$

and

$$r = k - p, \quad K = k \cdot p, \quad P = p \cdot q.$$

Take

$$A(k, p) = \frac{k_0 + m}{V_k V_{k-p}}, \quad C(p, q) = \frac{p_0}{V_p V_{p+q}}, \quad X = A C,$$

with

$$d_{\gamma} = 1, \quad d_{\Gamma} = 2.$$

Child operator

The ordinary child projection is

$$U_{\gamma,1}A = \frac{k_0 + m}{U_k^2} + \frac{2k_0 K}{U_k^3}.$$

The child soft projection is

$$S_{\gamma,0}A = \frac{k_0 + m}{V_k^2}.$$

Applying the ordinary UV projection to that completed soft term gives

$$U_{\gamma,1}S_{\gamma,0}A = \frac{k_0 + m}{U_k^2}.$$

Therefore

$$H_{\gamma}A = \frac{k_0 + m}{V_k^2} + \frac{2k_0 K}{U_k^3}.$$

This explicitly shows that the soft branch retains m , while M occurs only through an ordinary UV projection.

Generic parent operator

For

$$Z = \frac{J_{\text{phys}}}{V_{p+q}}, \quad J_U(u) = j_0 + u j_1 + u^2 j_2,$$

the parent UV projection is

$$U_{\Gamma,2}Z = \frac{j_0}{U_p} + \left(\frac{j_1}{U_p} - \frac{2Pj_0}{U_p^2} \right) + \left[\frac{j_2}{U_p} - \frac{2Pj_1}{U_p^2} + j_0 \left(\frac{4P^2}{U_p^3} - \frac{q^2 + \Delta}{U_p^2} \right) \right].$$

The parent soft projection is

$$S_{\Gamma,1}Z = J_{\text{phys}} \left(\frac{1}{V_p} - \frac{2P}{V_p^2} \right).$$

The overlap is

$$U_{\Gamma,2}S_{\Gamma,1}Z = \frac{j_0}{U_p} + \left(\frac{j_1}{U_p} - \frac{2Pj_0}{U_p^2} \right) + \left[\frac{j_2}{U_p} - \frac{2Pj_1}{U_p^2} - \frac{\Delta j_0}{U_p^2} \right].$$

Consequently,

$$H_{\Gamma}Z = J_{\text{phys}} \left(\frac{1}{V_p} - \frac{2P}{V_p^2} \right) + j_0 \left(\frac{4P^2}{U_p^3} - \frac{q^2}{U_p^2} \right).$$

Define

$$L_m = \frac{1}{V_p^2} - \frac{2P}{V_p^3}, \quad Q_U = \frac{4P^2}{U_p^4} - \frac{q^2}{U_p^3}.$$

Parent acting on the bare child

For the bare expression,

$$A_0 = \frac{k_0}{U_k U_r},$$

and

$$j_0^X = \frac{p_0 k_0}{U_k U_r U_p}, \quad j_1^X = \frac{p_0 m}{U_k U_r U_p},$$

$$j_2^X = -\Delta j_0^X \left(\frac{1}{U_k} + \frac{1}{U_r} + \frac{1}{U_p} \right).$$

Thus

$$H_{\Gamma}X = p_0 A L_m + p_0 A_0 Q_U.$$

Parent acting on the completed child

Let

$$\hat{A} = H_{\gamma} A.$$

Under the parent UV grading,

$$g_0 = \frac{k_0}{U_k^2} + \frac{2k_0 K}{U_k^3}, \quad g_1 = \frac{m}{U_k^2}, \quad g_2 = -\frac{2k_0 \Delta}{U_k^3}.$$

Then

$$H_{\Gamma}(\hat{A}C) = p_0 \hat{A} L_m + p_0 g_0 Q_U.$$

Four distinct forest terms

GammaLoop must retain these four signed forest terms separately:

$$\begin{aligned}
F_{\emptyset} &= +\frac{p_0 A}{V_p V_{p+q}}, \\
F_{\gamma} &= -\frac{p_0 \hat{A}}{V_p V_{p+q}}, \\
F_{\Gamma} &= -p_0 A L_m - p_0 A_0 Q_U, \\
F_{\gamma;\Gamma} &= +p_0 \hat{A} L_m + p_0 g_0 Q_U.
\end{aligned}$$

Only after each term has been matched to its forest provenance may tests regroup them:

$$\begin{aligned}
F_{\emptyset} + F_{\gamma} &= \frac{p_0 (A - \hat{A})}{V_p V_{p+q}}, \\
F_{\Gamma} + F_{\gamma;\Gamma} &= -p_0 (A - \hat{A}) L_m + p_0 (g_0 - A_0) Q_U,
\end{aligned}$$

and therefore

$$R = p_0 (A - \hat{A}) \left[\frac{1}{V_p V_{p+q}} - L_m \right] + p_0 (g_0 - A_0) Q_U.$$

The required asymptotic checks are

$$A - \hat{A} = O(k^{-5}), \quad A_0 - g_0 = O(k^{-5}),$$

the pure parent hard behavior is $O(p^{-5})$, the simultaneous two-loop hard behavior is $O(\Lambda^{-9})$, and

$$\frac{1}{V_p V_{p+q}} - L_m = O(q^2), \quad Q_U = O(q^2).$$

Run this derivation at generic $m \neq 0$ to prove mass preservation and at $m = 0$ to provide a simple soft-profile oracle. Paper formulas may combine several forest contributions. GammaLoop tests must first compare each forest family separately; in 3D, residues belonging to the same forest family may be summed, but raw terms from different families may not be pre-combined merely to match the paper's presentation.

6. Tests and fixtures

Algebraic operator tests

Add active normalized-atom tests for the soft portions of Eqs. 4.15, 4.16, and 4.19:

- $d = 0$: $H_0 = U_0$.
- $d = 1$: S_0 retains the constant soft term.
- $d = 2$: S_1 retains constant and linear soft terms.
- Physical masses survive in S .
- M appears only in U and US .
- No auxiliary Taylor parameter survives evaluation.
- Verify $1 - H = (1 - U)(1 - S)$ and $H - U = (1 - U)S$.
- Include a noncommuting example demonstrating that $US \neq SU$.
- Assert the marker and subtraction sign occur once, after branch combination.

Nested and disconnected tests

Cover the operator matrix

$$(U/U), \quad (H/U), \quad (U/H), \quad (H/H)$$

for child/parent choices:

- Retain all four signed forest terms.
- Check the child remainder in the child-hard limit.
- Check the paired outer terms in the child-hard limit.
- Check the complementary pair in the reduced-parent limit.
- Check simultaneous hard rays.
- Claim soft improvement only for a complete soft wood, not for an arbitrarily incomplete subset.
- Verify all branches use the same canonical route.
- Add an affine-routing negative test which produces a contextual error.
- Test disconnected soft components with HedgePoset and verify factorization into completed component results.

3D micro-oracle and ownership tests

For a one-time/one-space CFF surface, use

$$E_m(\ell) = \sqrt{\ell^2 + m^2},$$

$$\eta(s) = E_m(\ell) + E_m(\ell - sr_1) + sr_0.$$

Its soft coefficients are

$$s_0 = 2E_m, \quad s_1 = r_0 - \frac{\ell r_1}{E_m}, \quad s_2 = \frac{m^2 r_1^2}{2E_m^3}.$$

For the ordinary UV deformation, with $E_U = \sqrt{\ell^2 + M^2}$,

$$u_0 = 2E_U, \quad u_1 = r_0 - \frac{\ell r_1}{E_U},$$

$$u_2 = \frac{M^2 r_1^2}{2E_U^3} - \frac{\Delta}{E_U}.$$

For

$$R(z) = \frac{N_0 + zN_1 + z^2N_2}{S_0 + zS_1 + z^2S_2},$$

use the independent oracle

$$R_0 = \frac{N_0}{S_0},$$

$$R_1 = \frac{N_1}{S_0} - \frac{N_0 S_1}{S_0^2},$$

$$R_2 = \frac{N_2}{S_0} - \frac{N_1 S_1}{S_0^2} + N_0 \left(\frac{S_1^2}{S_0^3} - \frac{S_2}{S_0^2} \right).$$

Test:

- Equal CFF atoms originating on different shores.
- Nested ownership where the child surface changes but its co-graph does not.
- Ambiguous ownership producing an error rather than global rewriting.
- U , S , US , and H against residues of the corresponding 4D expressions.
- Nested 4D-to-3D residue compatibility.
- Exactly one final combined H atom per residue.

Paper and topology fixtures

Keep separately named fixtures for:

- The exact massive nested self-energy topology presented in Appendix B.1.
- The graph labelled B.1 in the paper's figure material, so it cannot be confused with the Appendix equation.
- A double triangle with a massless-quark loop.
- Its soft-IR variant.
- Bare massless fermion and gluon self-energies.
- The DGSE/figure-8-style soft self-energy example.
- Consecutive, nested, and disconnected soft components.

The exact massive Appendix-B.1 graph must parse, validate, identify its connected components, expose the expected forest provenance through inspect-level output, and reach the intentional OS panic if configured with the paper's OS assignment. A separately configured IR specialization must exercise the active soft implementation.

After canonical tensor/gamma normalization, compare the applicable local soft expressions with Appendix B.5–B.7, the massless B.10–B.12 specialization, and the nested B.16 result. Do not claim coverage of integrated expressions B.8–B.9, B.13–B.14, B.17 onward, or OS-only formulas.

Inspect-level tests must display or serialize:

- Selected connected components and their approximation types.
- Canonical loop routes.
- Parent/child forest keys.
- Separate U , S , and US provenance before their final combination.
- Physical versus UV-mass denominators.
- Residue and CFF-surface ownership.

No profile test may pass vacuously: selected and resolved limit counts must both be nonzero.

Actual UV-profile CLI acceptance

Every representative graph-level fixture must include a run-card command that is parsed and executed through the real CLI path:

```
profile ultra-violet
```

Constructing `Profile::UltraViolet` directly is insufficient as the only integration test.

Use summed-orientation mode unless a test explicitly targets orientation bookkeeping. For example:

```
profile ultra-violet \
  -p gamma_star_ddbar_top_bubble \
  -i soft_ir \
  --min-scaling 8 \
  --max-scaling 12 \
  --n-points 25 \
  --seed 1337
```

The command writes a profile output directory; acceptance must parse its `uv_profile.json`, not infer success from the process exit code.

For each locally subtracted fixture require:

- Total selected limits > 0 .
- Resolved fits > 0 .
- Finite fitted exponents.
- `pass_fail(-0.9).failed == 0`.

Run a matched unsubtracted control with the same topology, routing, seed, and scaling window. It must have selected and resolved limits > 0 and fail at least one expected UV limit.

Lightweight primitive and self-energy CLI profiles belong in the normal suite. Larger Appendix and vertex profiles may be marked serial/slow, but they remain mandatory in the project's acceptance CI and may not be silently ignored.

Physical $q_g \rightarrow 0$ massive-top-bubble vertex test

Add a two-loop vertex fixture named `gamma_star_ddbar_top_bubble`:

- External process: $\gamma^* \rightarrow d\bar{d}$, with massless external $d, \bar{d}$.
- Use q_g for the internal soft gluon momentum to avoid confusing it with the external quark label.
- Dress the exchanged gluon with a closed massive top-quark self-energy.
- The two massless gluon propagators on either side of the top bubble carry the same q_g .
- Give the corrected gluon a stable selector edge, fixed in the fixture as `e6`.
- Close external light-quark color with δ_i^j/N_c .
- Use nonexceptional fixed kinematics below the top-pair threshold, with $Q = \sqrt{p_{\gamma^*}^2} = 300$ GeV, $m_t = 173$ GeV, and zero top width.

Generate three otherwise identical local-only integrands:

1. `bare`: no local counterterms.
2. `uv_only`: ordinary U on every selected connected component.
3. `soft_ir`: H on the connected massive-top gluon self-energy and ordinary U on the overall vertex and other components.

The soft component has $d = 2$, so

$$H_2 = U_2 + S_1 - U_2 S_1.$$

At fixed top-loop momentum ℓ ,

$$(1 - S_1)\Pi_{\mu\nu}(q_g, \ell; m_t) = O(q_g^2).$$

Consequently, the repeated-gluon insertion changes from

$$I_U(q_g) \sim q_g^{-6}$$

to

$$I_H(q_g) \sim q_g^{-4}, \quad \frac{I_H}{I_U} \sim q_g^2.$$

This removes the spurious two-power enhancement but does not remove the genuine logarithmic soft behavior of the full massless vertex.

Run `profile ultra-violet` on all three variants. The bare variant must fail at least one hard limit; both `uv_only` and `soft_ir` must pass, proving that the soft refinement preserves ordinary UV subtraction.

Then run matched soft profiles:

```
profile bulk \
  -p gamma_star_ddbar_top_bubble \
  -i uv_only \
  --select 'top_bubble_vertex S(e6)' \
  --min-scaling -2 \
  --max-scaling -5 \
```

```
--n-points 25 \
--seed 1337
```

```
profile bulk \
-p gamma_star_ddbar_top_bubble \
-i soft_ir \
--select 'top_bubble_vertex S(e6)' \
--min-scaling -2 \
--max-scaling -5 \
--n-points 25 \
--seed 1337
```

Run the same soft profile for `bare` as a diagnostic control.

Acceptance requires:

- Exactly one requested soft report per variant.
- A resolved, finite fit with $R^2 \geq 0.98$.
- Identical route, random direction, seed, points, and fit window between variants.
- The UV-only fitted scaling is less than -1.0 , with theoretical expectation approximately -2 .
- The soft-refined fitted scaling is greater than -0.5 , with theoretical expectation approximately 0 .
- Most importantly,

$$\text{scaling}_H - \text{scaling}_U \geq 1.5,$$

with theoretical target $+2$.

GammaLoop's bulk profile includes the three-dimensional soft integration measure, which is why $q_g^{-6} \rightarrow q_g^{-4}$ appears approximately as $-2 \rightarrow 0$. Do not require `soft_ir.all_passed`: the remaining marginal behavior is physical.

This fixture is also the principal mixed-nesting test: the outer ordinary U must re-expand the complete child H result without destroying its $O(q_g^2)$ soft remainder.

7. Completion criteria and deferred work

Phase 1 is complete only when:

- OS dispatch is cleanly and intentionally unimplemented in both representations.
- No partial γ^0 , $P_\pm$, or rest-frame graph rewrite remains active.
- IR/soft plus integrated generation fails before expression generation.
- The 4D and 3D implementations satisfy $H = U + S - US$ for $d = 0, 1, 2$.
- Physical masses remain in soft branches and M arises only through ordinary UV projection.
- Nested U/H forests pass separate child, parent, and simultaneous asymptotic checks.
- The toy example's four forest terms match independently before regrouping.
- The Appendix-B.1 topology is a valid, non-vacuous GammaLoop fixture.
- CFF replacement is component-local and occurrence-safe, with ambiguous ownership rejected.
- Lightweight profile tests run in the normal suite and disconnected-union tests use HedgePoset.
- Representative fixtures execute the actual `profile ultra-violet` CLI command with nonzero selected/resolved limits and matched failing bare controls.
- The massive-top-bubble vertex passes ordinary UV profiling in both `uv_only` and `soft_ir` modes and demonstrates the expected two-power $q_g \rightarrow 0$ improvement.
- Formatting, targeted tests, and the applicable full test suite pass without weakening unrelated tests.

Phase 2 will integrate the single completed local soft atom under a scheme-specific policy and test $[S] = 0$ and the integrated Appendix-B formulas. No soft branch may be dropped or replaced by an integrated identity before nesting is complete.

Phase 3 will revisit OS counterterms after GammaLoop can derive local counterterms directly from the expanded 4D representation, where the nonzero rest-frame expansion point and γ^0 projectors can be introduced without rewriting the 3D CFF graph in place.

</proposed_plan>

Implementation finding: contracted-child grading in hybrid soft/MUV woods

The first numerical DGSE profile falsified an implementation inference made while realizing the ordered forest recurrence. Applying the same simultaneous full-parent Laurent grading to both the homogeneous all-MUV child channel and the finite soft refinement violates the contracted-graph semantics of the forest formula.

For a degree- d_γ soft-refined child, write

$$D_\gamma = H_{d_\gamma} - U_{d_\gamma} = (1 - U_{d_\gamma})S_{d_\gamma-1}.$$

Write the completed child as $H_\gamma = U_\gamma + D_\gamma$. The soft jet decomposes D_γ into child-boundary polynomials of degree $j \leq d_\gamma - 1$, with child-UV-finite coefficients after the factor $(1 - U_{d_\gamma})$. When such an insertion is viewed in a containing component of superficial degree d_Γ , the reduced-parent degree of its order- j term is

$$d_\Gamma - d_\gamma + j \leq d_\Gamma - 1.$$

Equation 3.3 of the paper represents the inner counterterm as an effective vertex of the contracted graph Γ/γ . The nested-cancellation conditions in Eqs. 3.7–3.8 then scale only the quotient loops while holding the completed child's internal loops fixed. The discussion following Eq. 4.5 makes the same requirement for the cancellation of a nested forest pair. In three dimensions the corresponding Laurent measure is therefore $\lambda^{3L_{\Gamma/\gamma}}$, not λ^{3L_Γ} , on the D_γ channel.

Linearity gives the required ordinary-parent action

$$U_\Gamma(H_\gamma * \Gamma/\gamma) = U_\Gamma^{\text{full}}(U_\gamma * \Gamma/\gamma) + U_\Gamma^{\text{quot}}(D_\gamma * \Gamma/\gamma).$$

The superscripts distinguish only loop-coordinate and loop-measure grading. The quotient operation still acts on the complete D_γ expression: it scales a child boundary carrier when that carrier is a quotient-loop momentum, grades physical masses and free M -numerator factors under the outer U , and preserves terminal- M denominator provenance. Only the child's internal loop coordinates and their measure are held fixed. For an H parent the same split is used in its U and US branches, while its S branch acts on the exact completed sum.

This is not a replacement of H_γ by U_γ , nor a dropped forest term. The implementation carries both channels explicitly and recombines them into the single signed parent atom. A logarithmic parent has no quotient projection of D_γ , because every term has degree at most -1 . For a positive-degree parent, U_Γ^{quot} subtracts any non-negative reduced-parent coefficient. A degree-zero soft child remains trivial because $H_0 = U_0$. Focused tests must verify both the reduced-parent and simultaneous hard limits before the positive-degree hybrid is accepted.

The exact DGSE failure exposed the distinction: with the child loop held fixed, the old two-loop three-dimensional grading inserted one extra loop measure and shifted the predicted degree from -1 to $+2$. The all-MUV reference channel already had the correct cancellation and remains unchanged.

Implementation finding: free versus terminal UV-mass factors

The sentence in the verbatim plan saying that a literal M remains fixed is too broad. Section 4.1 of arXiv:2203.11038v1 states after Eq. (4.4) that an outer Taylor operation must also expand the m_{UV} factors which an inner Taylor operation introduced in a numerator. Consequently, a free inner-counterterm numerator factor m_{UV}^r has outer- U Taylor degree r , just like any other polynomial mass factor.

This is distinct from a terminal UV-vacuum denominator. Such a denominator retains its terminal provenance and vacuum-denominator shape under an outer operation: it must not be reinterpreted as a physical-mass propagator and subjected to a second UV rearrangement. The implementation preserves this role distinction in both representations:

- physical masses and free numerator mass factors are graded once by an ordinary outer U ;
- a soft S operation leaves all masses unchanged;
- terminal M -denominators remain terminal and are graded only through the homogeneous scaling of the completed local expression.

The corresponding direct oracle must therefore distinguish a free inner- U numerator factor from the mass embedded in a terminal denominator; using one symbol at Taylor degree zero to represent both roles would contradict the paper.

Implementation finding: mixed CFF occurrences require a partial image

The verbatim plan's instruction to leave a mixed-boundary surface unchanged must be understood as forbidding a cache-global replacement or a deformation of its pure co-graph contribution. It cannot mean leaving the concrete mixed occurrence algebraically unchanged. Taking the nested energy kernel

$$I_s = \frac{1}{(k_0^2 - A^2)((k_0 - s\ell_0)^2 - B(s)^2)(\ell_0^2 - C^2)}$$

at the matched poles $k_0 = A$ and $\ell_0 = C$ produces

$$\text{Res } I_s = \frac{1}{4AC [A - sC - B(s)] [A - sC + B(s)]}.$$

The standalone co-graph residue factor $1/(2C)$ is fixed. Inside each mixed CFF denominator, however, the same C represents the temporal momentum entering the active child and must therefore acquire the soft parameter. If

$$B(s) = \sqrt{(k_1 - s\ell_1)^2 + m^2},$$

then at $s = 0$

$$\frac{d}{ds}[A - sC - B(s)] = -C + \frac{k_1\ell_1}{B(0)}, \quad \frac{d}{ds}[A - sC + B(s)] = -C - \frac{k_1\ell_1}{B(0)}.$$

Freezing the mixed occurrence would miss the $-C$ term, while globally scaling its cached denominator would also deform pure co-graph occurrences and energy prefactors. The occurrence-local `Partial` image is therefore the required residue map: prepare the component-internal energy dependence and scale only the non-component-energy plus external-shift combination which represents the active component's boundary energy. Pure co-graph occurrences remain unchanged, and equal algebraic denominators on different shores retain separate occurrence histories.

Appendix-B v1 traceability notes

The implementation audit found two internal inconsistencies in the literal v1 Appendix which must not be mistaken for GammaLoop routing or operator choices.

First, B.3 prints

$$\gamma_2 = \{e_1, e_2, e_4, e_5\},$$

and the surrounding B.5 text describes the quotient as a massive tadpole. That printed set is not the 1PI fermion cycle in the displayed topology. The topology itself and the surviving $(k - p)^2$ cograph in B.20 instead select $\{e_1, e_2, e_3, e_4\}$, leaving the massless paper edge e_5 . In the checked-in GammaLoop graph the internal-edge map is

$$e_2^{\text{GL}} \mapsto e_2^{\text{paper}}, \quad e_3^{\text{GL}} \mapsto e_4^{\text{paper}}, \quad e_4^{\text{GL}} \mapsto e_5^{\text{paper}}, \quad e_5^{\text{GL}} \mapsto e_3^{\text{paper}}, \quad e_6^{\text{GL}} \mapsto e_1^{\text{paper}}.$$

GammaLoop therefore correctly uses the 1PI child $\{e_2^{\text{GL}}, e_3^{\text{GL}}, e_5^{\text{GL}}, e_6^{\text{GL}}\}$ and leaves $e_4^{\text{GL}} = e_5^{\text{paper}}$ in the cograph. Fixture comments and test names record this explicitly instead of claiming an unqualified literal B.5 edge assignment. Second, the first quadratic numerator in the TeX of B.16 has a sign which, read literally, conflicts with both B.7 and direct second differentiation of the B.1 integrand. The implementation uses the direct-expansion sign $-q^2 N_0/U_p^4$, and its test records the v1 discrepancy before comparing the physical and UV products as separate forest families.

Corrected implementation finding: the outer forest operator remains the full parent operator

The earlier “contracted-child grading” note above records an intermediate diagnosis which is superseded by the explicit DGSE power count and CFF trace below. Contracted-graph scaling is a useful *diagnostic limit* in the nested cancellation proof, but it is not a second production definition of K_Γ . The forest recurrence remains

$$K_\Gamma(1 - K_\gamma)I,$$

with the one full parent operator acting linearly on the completed child remainder. In particular, there is no scheme-dependent split in which the all- U part is graded with all parent loops while $H_\gamma - U_\gamma$ is graded with quotient loops only. For the DGSE fixture, the massless two-edge child has $d_\gamma = 1$, while the containing graph is logarithmic, $d_\Gamma = 0$. At fixed child loop momentum k ,

$$D_\gamma = H_{\gamma,1} - U_{\gamma,1} = S_{\gamma,0} - U_{\gamma,1}S_{\gamma,0}$$

is independent of the quotient loop momentum q . The reduced cograph has four propagators and a numerator of rank three in q , so

$$D_\gamma C_{\Gamma/\gamma} = O(q^{-5}),$$

or degree -1 after the four-dimensional loop measure. Its quotient hard projection is therefore zero, exactly as the nested forest recurrence requires for a logarithmic parent. Consequently the DGSE H/U and U/U outer-node atoms agree, even though the child $H - U$ atom is nonzero. Tests must establish both facts non-vacuously.

Corrected implementation finding: the soft CFF jet must use factorised residue blocks

The erroneous DGSE three-dimensional scaling was not caused by a missing outer counterterm. For one contributing orientation the unexpanded full CFF contains the four mixed causal factors

$$[0, 5, 6], \quad [4, 5, 6], \quad [1, 5, 6], \quad [5, 6, 8].$$

Applying the occurrence-local partial image at the exceptional child-soft point sends all four denominators to the same child energy sum $E_5 + E_6$. The result contains $(E_5 + E_6)^{-4}$, whereas the residue of the Taylor-expanded four-dimensional graph contains only one child causal factor and three causal denominators of the contracted cograph. Losing those three quotient denominators shifts the observed hard degree by precisely three powers, from the required -1 to $+2$.

Thus the earlier mixed-occurrence note is locally correct as a derivative of one generic, nonexceptional denominator, but it is insufficient as a definition of the *completed* CFF soft jet: taking the exceptional soft Taylor jet and resolving the energy residues is non-uniform because poles coalesce. The production source for a soft branch must instead be the fixed-orientation factorisation

$$\text{CFF}(\delta/\{\text{immediate soft children}\}) \prod_{\gamma < \delta} \text{CFF}(\gamma/\{\text{its immediate soft children}\}) \text{CFF}(\Gamma/\delta),$$

with the corresponding numerator kernels and with the full-orientation selector inserted exactly once. A soft map acts on the active component block and its numerator while the pure cograph block remains literal; an outer operation subsequently acts on the actual completed product. Nested and disjoint soft components therefore form a laminar block forest. Equal denominators retain mathematical equality, while block/occurrence provenance records where each factor came from.

The same CutCFFIndex and canonical full orientation must be retained while the block inventory is allowed to differ between U , S , and US . Branch materialisation must consequently validate a union of branch-tagged occurrences rather than require byte-for-byte identical occurrence tables. If a selected residue cannot be transported to the factorised block forest unambiguously, generation must fail with graph, component, route, residue, orientation, and block context; it must never revert to a global surface replacement.

Corrected implementation finding: Phase-1 hybrid closure is degree dependent

The unconditional hybrid-wood allowance in the verbatim plan is too broad. Let a positive-degree child use (H_γ), and isolate its finite scheme refinement ($D_\gamma = H_\gamma - U_\gamma$). A boundary Taylor monomial of order ($j_d_\gamma - 1$) has reduced-cograph degree

$$d_{G_\gamma/\gamma}(D_\gamma)_j = d_{G_\gamma} - d_\gamma + j.$$

The worst retained monomial therefore has degree ($d_\gamma - 1$). A logarithmic ordinary parent ($(d_\gamma = 0)$) is safe: its quotient projection vanishes, as the explicit DGSE power count above shows. For an ordinary MUV parent with ($d_\gamma > 0$), however, the same hybrid leaves a nonnegative reduced-cograph limit. Two independent 4D fixtures expose the limiting degree zero for ($d_\gamma = 1$), even though the full-parent recurrence is implemented exactly.

Replacing the full parent operation by a quotient-graded operation on (D_γ) is not a valid repair: it contradicts the Appendix-B.16 completed-child expression and makes the production operator depend on an arbitrary algebraic split. The missing object is a finite scheme-change counterterm after the child has been integrated to a local vertex, which is part of the deferred integrated policy. Phase 1 must consequently allow (H/U) only for ($d_\gamma = 0$) and reject a positive-degree ordinary-MUV ancestor of a retained soft refinement with a contextual error. Same-scheme (H/H) nesting remains active, as does the logarithmic DGSE (H/U) case.

Corrected implementation finding: ordinary production and soft replay are separate 3D channels

Ordinary U continues to act on the established full CFF production atom. Replacing that atom by a product of generated component/cograph blocks at a generic hard point changes the residue representation and spoils ordinary UV cancellation. In parallel, each deferred term retains a laminar factorised replay state for a possible later S operation. Ordinary U updates that replay with the same Taylor parameter and route but does not publish it as the production atom. Only S consumes the replay and reconstructs the production atom from its blocks.

Each replay block therefore carries its own transformed-occurrence history. An outer S must freeze the actual image left by an inner U , S , or US , rather than cancel its tag with the original physical denominator; otherwise a spurious ratio such as $(D/U(D))$ survives and the ordered jet is lost. Replay provenance becomes public production provenance only when S consumes the factorisation. Splitting a previously deformed container is rejected if its ordered occurrence images cannot be transported unambiguously.

Corrected implementation finding: only soft components create CFF block boundaries

The fixed-orientation laminar factorisation is an exceptional-point replay program, not an alternative value for a generic ordinary counterterm. In particular, for an ordinary child $\gamma \subset \Gamma$ there is no orientation-wise identity

$$B_{\Gamma}^{(o)} = B_{\gamma}^{(o)} B_{\Gamma/\gamma}^{(o)}.$$

Consequently an ordinary U_{γ} operation must deform the smallest existing causal container in place and record its unprojected deformation history; it must not introduce a new component/cograph split. Only a soft operation creates the block boundary required when its exceptional limit coalesces poles. The replay root is the maximal internal causal region: external half-edges are not part of that region and must not force an overall soft component through a spurious split.

If a later proper soft component splits a container carrying earlier operations, occurrence images from the discarded causal tree cannot simply be copied to the regenerated trees. There is generally no occurrence bijection, and an operation may have changed inverse-energy or tree factors even when it emitted no surface event. Instead, retain the exact ordered, unprojected deformation program. For each regenerated block, replay the inherited stages in chronological order before appending the new soft stage, while leaving the already-deformed numerator/measure kernel untouched. A stage represented by an already contracted child block is not replayed on its cograph.

This ordering follows from the ring-homomorphism property of the deformation,

$$\Phi(K \prod_i C_i) = \Phi(K) \prod_i \Phi(C_i),$$

whereas the Taylor projector itself is not multiplicative:

$$J(fg) \neq J(f)J(g).$$

The formal Taylor parameters therefore remain shared across the regenerated factors and are projected only after their product has been recombined. The same replay representation is required for inherited ordinary and soft stages; special-casing only U/H would silently mis-order H/H woods. Any non-laminar scope or route which cannot be replayed exactly remains a contextual generation error rather than falling back to global replacement.

Implementation finding: selected CFF residues require local count provenance

A selected `CutCFFIndex` records a residue order, but it does not identify the raised surface group which was selected. The factorised replay therefore retains the originating `CutSet` and assigns a local right-threshold, left-threshold, and LU count to every causal block. `Some(0)` is meaningful internally: the selector exists globally, while that particular factor must contain no occurrence. Branch selection is performed in the same right-to-left-to-LU order as `Graph::cff`, and a selected denominator is mapped to the residue unit only after the matching branch count has been enforced.

Pruning a causal tree must not renumber away occurrence provenance. A `NormalizedCFFBlock` consequently stores the complete occurrence object in each tree node, rather than recovering it from the post-pruning node index. This preserves the original occurrence identity, shore, and support even when equal denominators occur on different branches.

The currently proven transport is deliberately conservative. An already selected complete block can be transformed directly. When a block is split around a component, every regenerated occurrence is first aligned with the orientation-filtered occurrence table of its source container. The key is the block-local generated shore, compared with the source shore after restriction to the visible region and modulo exchange with its complementary component shore, together with signed energy support. A source on the same restricted shore is preferred; use the complementary shore only when no same-shore source matches. Within that class, when several global shores have the same component restriction, compare their sets of source-shore vertices outside the component by inclusion. Discard only strict supersets; retain incomparable minimal extensions, which therefore remain an ambiguity error unless they give the same exact copied image. Cardinality is not a valid topological order for branching cographs. For the freshly generated component block, the source energy support is projected onto the exact visible uncontracted domain `paired_edges(region - generator_contract)` before comparison. This is only a matching key: after the unique match, its effective surface ID, complete denominator, and unprojected support are copied from the source while the generated surface ID, local shore, and stable node ID remain block-local provenance. For example, the top-bubble block generates support $E_7 + E_8$, while the fixed-orientation full-CFF source shore carries $E_5 + E_7 + E_8$; projecting the source key identifies the component shore, but retaining its full support lets the soft map form $E_7 + E_8 + sE_5$. Cograph matching remains exact

and unprojected, since projecting a cograph source could copy a contracted child's energies back into the quotient. After topological ranking, source contexts which give exactly the same copied image – equal surface ID, full support, and algebraically equal denominator – are one operational remap candidate: source context is not copied, and the generated tree keeps its own occurrence node and multiplicity. Two distinct full images at the same best rank remain an ambiguity error. In particular, regenerated and source node indices are not used as a tie-break because their causal trees can have different local numbering. The block atom is then rebuilt before any residue selection. This shore lift is essential: contraction can erase a literal external-energy shift such as $\pm Q_5^0$, and no later soft boundary map can reconstruct a symbol which is already absent. Copying the unique source occurrence leaves a pure cograph denominator literal while still allowing the active component's soft map to scale its own copy of the boundary shift. The freshly aligned component/cograph blocks are the residue representation of the corresponding *soft forest branch*; they are not an undeformed identity factorisation of the generic root CFF at ($s=1$). The top-bubble orientation zero is an explicit counterexample: its full graph orientation is acyclic and has a nonzero root-CFF atom, but contracting the top bubble makes the inherited cograph orientation cyclic. The component block is nonzero and the contracted cograph block is therefore zero. Requiring their product to reproduce the generic root term would compare two different forest graphs and reject the mathematically correct residue representation. The top regression records this counterexample explicitly, while the exact operation-before-residue tests compare the (S), (US), and (H) branches with their corresponding 4D residues. The mandatory summed-orientation top-bubble profile supplies the full-graph physical acceptance. After this lift, a selected count may be assigned wholly to the cograph only if the prospective zero-count component block contains no occurrence of the selected raised surface. Exact support/denominator matching is preferred; an energy-support fallback is allowed only when no global surface outside the selected raised group has the same energy multiset. A positive local count with no match is an error, as is a zero local count with a match, because the latter requires residue-count convolution. All physical cut edges are passed to the regenerated block. If the selected cut intersects the component, a source shore has zero or multiple lifts, or a selected-equivalent occurrence appears in a block provisionally assigned count zero, Phase 1 rejects the operation with graph, component/block, residue, orientation, cut, and route context rather than generating an unselected or shift-erased CFF.

Superseded implementation finding: fixed-orientation provenance versus numerical profiles

An earlier diagnostic proposed selecting one stored orientation inside the computed-inspect exporter and remapping a dense occurrence index back to that stored index. That selector was not retained: graph terms already contain the orientations selected during generation, and no export consumer selected a further orientation. Computed export therefore consumes the graph term's existing orientation set without another filtering or indexing convention. Actual orientation-local validation selects orientations through the normal generation/profile path, while summed-orientation UV and bulk profiles remain the additional physical cancellation checks.

The large top-bubble fixture uses a computed HedgePoset 4D forest export for its graph-specific parent/child, route, and U/S/US/H provenance assertions. That export integrates only the recursive 4D forest, emits one all-none residue atom per node, and deliberately bypasses every three-dimensional $(2\pi)^{3L}$, LU, and CFF post-factor. Its actual generated integrands and all UV and q_g -soft CLI profiles remain three-dimensional and summed over orientations. Smaller exact 3D residue oracles retain the detailed CFF route, mass-wrapper, occurrence-ownership, and nested-replay checks. This separation avoids recomputing and serializing a giant exact 3D provenance atom merely to repeat structure already checked by those oracles.

Generation RAM telemetry polls at one-second cadence. `sysinfo 0.33` scans the host `/proc` inventory even for a single requested PID, so the previous 100-ms cadence consumed a substantial fraction of one CPU during exact Symbolica work. The monitor parks with a timeout and is explicitly unparked before joining, preserving immediate generation shutdown despite the longer sampling interval.

Profile JSON is itself part of the acceptance boundary. The serialized form therefore retains `allow_vanishing_missing_fits`, aggregate inspect-fit status, and per-orientation inspect-fit status. `pass_fail` reads those public serialized entries after a round trip, so parsing `uv_profile.json` has the same missing-fit and vanishing-limit semantics as the in-memory result.

Latest implementation status: production deforms the summed unsplit CFF

This finding supersedes the later claims above that a soft branch should use a fixed-orientation component/cograph factorisation, that `S` should publish such a replay as its production atom, or that blockwise fixed-orientation identities are acceptance criteria. Those constructions may remain useful for topology and occurrence-provenance diagnostics, but they are not the mathematical CFF integrand on which the production Taylor operator acts.

The physical CFF is the sum over compatible orientations. Contracting a soft component does not preserve acyclicity orientation by orientation: an acyclic full-graph orientation can induce a cyclic contracted cograph, even though its term in the unsplit full CFF is nonzero. Contributions therefore redistribute and cancel between orientations, and in general there is no identity

$$\text{CFF}^{(o)}(\Gamma) = \text{CFF}^{(o)}(\delta) \text{CFF}^{(o)}(\Gamma/\delta)$$

for each fixed orientation o . Replacing an unsplit term by that product changes the residue representation before the Taylor jet is taken. Production must instead apply the component-local soft deformation to the complete summed, unsplit CFF atom, retain occurrence provenance only to distinguish equal owned and unowned factors, and project the Taylor series after that complete atom has been deformed. Fixed-orientation block tests must consequently be labelled as diagnostic construction tests; they must not require root/block equality, branch cancellation, or operation-before-residue identities orientation by orientation. Physical residue comparisons and profile acceptance are summed over the matched orientations.

The massive-top-bubble child profile is the numerical acceptance for this ordering. On the same route, seed, direction, and fit window, ordinary U_2 scales approximately as -2 , while the summed unsplit $H_2 = U_2 + S_1 - U_2 S_1$ result scales approximately as 0 . The observed two-power improvement is the required cancellation; differing individual orientation scalings are not a replacement for this summed physical result.

The canonical stored spatial-vector form is `Q3(edge, minkowski_representation)`, constructed as `GS.emr_vec(edge, Minkowski{}.new_rep(4).to_symbolic([]))`. An explicitly indexed component may use `Q3(edge, cind(mu))`, but a typed bare `Q3(edge)` is not a second convention. Replacement and routing code must match `Q3(edge, W_.x__)` and preserve its second argument so represented tensors and explicit components follow one deformation without losing their structure.

Authoritative correction: exceptional residues require conditional causal factors

This section supersedes the immediately preceding “Latest implementation status” section. It does not alter the verbatim implementation plan at the start of this file.

The summed, unsplit CFF remains the correct production representation for an ordinary U operation at generic kinematics. It is not, however, the correct object on which to take the component-local exceptional soft jet after the residue sum has flattened distinct child and co-graph poles. In the massive-top-bubble test, three mixed child/co-graph causal denominators coalesce with the child denominator at the child soft point. Taylor expanding that collapsed atom produces a fourth power of the child denominator and loses the three quotient denominators. This is precisely the observed spurious $+2$ scaling in five reduced-parent hard rays. Occurrence tags can distinguish equal factors, but cannot recover causal factors which have already coalesced algebraically.

The correct exceptional residue is a *conditional* causal factorisation; it is not the rejected construction in which one original full-graph orientation is imposed on every factor. For each quotient orientation o_Q , generate the contracted quotient first. Copy its directed boundary edges into the split edges of the child, independently sum every acyclic child-internal orientation compatible with that boundary, apply the child operation to that completed child sum, and only then multiply the untouched quotient factor:

$$\text{Res}(S_\delta I) = \sum_{o_Q} \Theta_Q(o_Q) C_Q(o_Q) S_\delta \left[\sum_{o_\delta | \partial o_\delta = \partial o_Q} \Theta_\delta(o_\delta) C_\delta(o_\delta) N_\delta \right].$$

The same ordering defines US : finish the soft Taylor polynomial of the child factor first, apply the ordinary child U operation to that completed factor, and multiply the quotient only afterwards. The ordinary U branch continues to use the established unsplit CFF path. A containing operator must replay the complete factorwise S or US branch from retained block and forest history; collapsing it and globally guessing ownership is invalid.

Selectors are owned exactly once. Original paired edges internal to the child contribute child selectors; quotient edges, including the child’s boundary carriers, contribute quotient selectors. Split boundary directions condition child

generation but do not contribute a second selector. A conditionally valid child sector need not lift to any single globally acyclic root orientation. Such a sector must be retained with generated factor-local occurrence provenance; source-CFF orientation matching is used only to align a unique occurrence image when one exists, never to pair or prune causal factors.

The top-bubble oracle establishes the concrete structure: the one-loop quotient has $2^4 - 2 = 14$ acyclic orientations, each has two conditioned child orientations, and every product contains exactly three quotient causal denominators and one child causal denominator. The inverse-energy factors partition edges $e3, \dots, e6$ and $e7, e8$ exactly once; there are no internal bridge/tree factors. Reversing the two boundary causal arrows alone does not send q_g to $-q_g$. The physical reversal combines the matched boundary-energy sectors with reversal of the canonical spatial carrier $Q3(e6, \text{mink}(4))$. Under that correlated map the selector-free S_0 coefficient is even and S_1 is odd.

Selected residues are convolved across factors. For every total `CutCFFIndex` order, enumerate non-negative child and quotient allocations whose componentwise sum is the requested right-threshold, left-threshold, and LU order. Select occurrences inside each factor before multiplication, retain selected images and factor-path occurrence namespaces, and reject ambiguous source alignment or overlapping raised groups contextually.

Finally, the single spatial-momentum convention is a two-argument function: $Q3(\text{edge}, \text{mink}(4))$ for a represented spatial vector, $Q3(\text{edge}, \text{mink}(4, \text{index}))$ for its automatically indexed tensor-network image, or $Q3(\text{edge}, \text{cind}(i))$ for an explicit component. Bare $Q3(\text{edge})$, arbitrary second arguments, and non-four-dimensional Minkowski representations are invalid. Replacement patterns use the exact one-slot form $Q3(\text{edge}, W_x_)$; $W_x_$ is not a supported compatibility convention.

Authoritative implementation-state addendum: completed conditional stages and one momentum representation

This section supplements the “Authoritative correction: exceptional residues require conditional causal factors” section and has precedence over every earlier post-plan implementation finding where they disagree. Historical findings remain in this file for traceability.

The ordinary U branch at generic kinematics continues to use the established summed, unsplit output CFF. Every branch containing an exceptional soft operation S , including US and a containing operation replaying a completed child S or US branch, uses the quotient-first conditional causal-factor tree. For each quotient orientation, first sum all acyclic child orientations compatible with its directed boundary, apply the connected child operation to that complete child sum, and only then multiply the untouched quotient factor. No single full-graph orientation is imposed on both factors. The output `CutCFF` determines the emitted residue-key domain. The separate unpruned, default-orientation alignment `CutCFF` is used only for occurrence provenance and raised-surface residue classification. A generated conditional factor and its generated denominator are always the mathematical production object; an aligned source occurrence never replaces that denominator, supplies a causal branch, or prunes a conditionally valid child sector. Occurrence identity includes the output residue, local residue allocation, factor path, and generated local occurrence namespace. Equal mathematical denominators remain equal. Requested right-threshold, left-threshold, and LU orders are convolved additively over child and quotient factors.

Every connected deformation stage is a completed sequential operator. If A_0 is the complete active-factor expression, then

$$A_i = \text{ev}_{z_i=1} J_{z_i}^{<n_i} \Phi_i(z_i) A_{i-1},$$

where $n_i = d_i$ for U and $n_i = d_i - 1$ for S . The parameter z_i is shared by the numerator, inverse-energy factors, causal denominators, tree factors, and every compatible-orientation term belonging to that active factor. The Taylor projector therefore acts on their complete sum and product, not on individual denominators or orientations.

After that projection, A_i contains no live z_i before stage $i + 1$ begins. In particular, USX means that the complete S Taylor polynomial is constructed first and a fresh U acts on that polynomial; it is not a joint bivariate Taylor projection. Stored jet entries are chronological replay recipes only. A retained pre-series occurrence image may be kept as diagnostic provenance, but it is not the physical atom on which a later operator acts. Conditional reconstruction replays one recipe, completes its Taylor jet at the active factor boundary, and only then advances to the next recipe or multiplies the factor into its quotient.

There is one canonical homogeneous full-route chart for the CFF kernel and its numerator. It is selected from the active connected component and its independent soft boundary carriers; an incoming child chart may be validated and converted, but it does not select a different parent chart. If $\mathcal{C}_r[N]$ is the numerator prepared in that ordinary- U full-route chart and q_a are the independent soft carriers, then

$$\Phi^S(s)N = \mathcal{C}_r[N]|_{q_a \mapsto sq_a}.$$

Dependent internal routes already expressed through $\mathcal{C}_r$ are not solved or substituted a second time. Carrier substitutions must not revisit the right-hand sides of those routed momenta, since that would introduce a spurious second power of s . Consequently,

$$\Phi^S(1)N = \mathcal{C}_r[N]$$

exactly. The massive-top-bubble numerator verifies this before tensor/gamma normalization in the canonical full route $[e6, e8]$, with $e6$ as its independent carrier. This equality is a chart invariant only: it does not by itself prove $SU = US$. A complete comparison must first route every dependent edge representative into the same homogeneous chart. In the top-bubble factor, $e5$ and $e6$ are opposite representatives of the same boundary momentum; comparing them as independent symbols produces a spurious nonzero $SU - US$ remainder. After routing both to $[e6, e8]$, the production soft remainder, the physical numerator, and the independent k^2 , k , q , and q^2 oracles vanish exactly through the requested soft order. This canonical equality must not be replaced by a second numerator-routing convention.

$Q3$ likewise has one structural convention: it is always a function of exactly two arguments. Its second argument is one representation object:

- $Q3(\text{edge}, \text{mink}(4))$ for the represented unindexed spatial vector;
- $Q3(\text{edge}, \text{mink}(4, \text{index}))$ for its automatically indexed tensor-network image;
- $Q3(\text{edge}, \text{cind}(i))$ for an explicit Cartesian component.

Thus these are different values of the same second slot, not different $Q3$ arities or conventions. Bare $Q3(\text{edge})$, arbitrary second arguments, and non-four-dimensional Minkowski representations are invalid. Replacement code uses exactly $Q3(\text{edge}, W_.x_); W_.x_$ is not a compatibility path.

Finally, ct_marker is unit-valued forest-history metadata, not part of the physical deformation algebra. Each deferred sum or active sector retains one accumulated marker history beside its marker-free physical terms. Neither Φ^U , Φ^S , a Taylor projector, conditional reconstruction, nor occurrence replay may transform that marker history.

Materialization first forms the signed marker-free $U + S - US$ result. Only after branch combination and the forest subtraction sign are complete is the current approximation marker attached once and the accumulated history updated. Public selected integrands and their ordinary-reference channel carry the same history, while conditional sources, jet recipes, and occurrence images remain marker-free. Disconnected products multiply their accumulated histories. Stripping markers must recover exactly the marker-disabled physical atom.

Authoritative implementation-state addendum: tree-bridge boundary flow

Conditional factorization distinguishes causal selector ownership from the direction needed to interpret a split child boundary. If a factor owns an ordinary tree bridge, that bridge carries no causal denominator and its two selector sectors have already combined through

$$\theta(q^0) + \theta(-q^0) = 1.$$

It is therefore contracted during that factor's CFF generation and contributes its ordinary propagator exactly once. The contraction records the bridge as undirected in the generated block, but a child meeting the same edge still requires a direction for its external-energy shift. Before recursing, restore the graph's deterministic `Orientation::Default` flow only on tree edges owned by the current factor. Both disjoint children then see the same global bridge flow, while opposite half-edge incidence supplies the appropriate local sign. This routing convention introduces neither a selector nor a second orientation sum.

Do not default arbitrary undirected split edges and do not relax the normalized CFF generator's missing-boundary-direction error. A non-tree split edge must still inherit its direction from the containing causal quotient; otherwise the conditional factorization is incomplete and must fail contextually.

Authoritative implementation-state addendum: conditional evaluator domain

This section has precedence over any earlier implementation note suggesting that exceptional (S) and (US) branches can remain on the flattened summed CFF through numerical evaluation. The ordinary (U) branch remains on that established representation. A branch containing (S) must instead use the quotient-first conditional residue representation described above.

The distinction is forced by the DGSE reduced-parent hard limits. For the degree-one child with internal edges (e5,e6), the four mixed supports

$$[e0, e5, e6], \quad [e4, e5, e6], \quad [e1, e5, e6], \quad [e5, e6, e8]$$

have the correct partial soft images

$$D_i(s) = E_5 + E_6 + sB_i.$$

Applying (S_0) only after those factors have been flattened produces $(E_5 + E_6)^{-4}$, removes three quotient denominators, and changes five reduced-parent hard rays from approximately (-1) to exactly (+2). Freezing the mixed factors is not a valid repair: it omits the boundary derivative required by the matched four-dimensional residue and destroys the top-bubble soft cancellation. The conditional residue instead contains one child denominator and three untouched quotient denominators.

A conditionally acyclic product need not lift to a globally acyclic root-CFF orientation. Such a product is nevertheless a physical term of the local counterterm and must be included in the numerical orientation sum. The conditional factor tree therefore exposes a separate evaluator-support sideband. For each complete product, merge only the directions on every factor's `owned_selector_edges`. These edge sets must be disjoint and must partition all paired non-tree, non-initial-cut selector edges. In the full signature, initial-state-cut edges use their fixed default direction and all other non-selector, tree, bridge, and unpaired edges remain undirected. Exact signatures are deduplicated. Occurrence namespaces are never interpreted as evaluator orientation IDs.

The root CFF orientation list remains the sole basis for CFF generation, threshold geometry, tropical sampling, public orientation labels, and explicit root-orientation selection. Only the parametric evaluator for the affected local atom receives the stable union

$$\mathcal{O}_{\text{eval}} = \mathcal{O}_{\text{root}} \cup \mathcal{O}_{\text{conditional}}.$$

Root orientations retain their existing indices as the prefix of that union. Threshold evaluators which do not contain an exceptional local branch continue to use the root list. A per-residue local or threshold atom which does contain such a branch receives its own conditional support. Summed evaluator modes must include the full union. Sampling or requesting one root orientation must fail contextually while non-liftable supplemental sectors are active, because there is no canonical one-to-one attribution of those sectors to a root orientation.

The massive-top-bubble factor supplies the structural oracle: its fourteen quotient alternatives and two conditioned child alternatives yield exactly twenty-eight complete selector signatures, with (e3,e8) directed and every other edge undirected. Numerical acceptance requires both sides of the representation issue: the top-bubble (q_g) fit improves by two powers, and all five DGSE reduced-parent hard rays return below the UV pass threshold.

Authoritative correction: complete conditional products use the root domain

This section supersedes the preceding claim that a completed conditional factor product can require a non-root evaluator orientation, and therefore supersedes the residue-wide evaluator-domain union proposed there.

The public CFF contracts only graph-theoretic tree bridges, not every edge outside the chosen loop-momentum basis. The massive-top-bubble topology is bridgeless: its ordinary root CFF consequently has thirty fully directed, globally acyclic orientations on (e3,e8). The conditional factor tree has fourteen acyclic quotient choices and two acyclic child choices. Its twenty-eight completed products are a strict subset of those thirty root orientations, rather than supplemental orientations.

More generally, under the validated laminar factor-tree construction, every complete conditional product lifts to a globally acyclic root orientation. If its merged directions contained a directed cycle, choose the lowest factor containing that cycle. The cycle either lies wholly in one child, contradicting the inductive hypothesis, or survives contraction as a directed cycle in that factor's quotient, contradicting the quotient CFF's acyclicity. Boundary directions are inherited consistently, and graph-theoretic tree bridges remain undirected in both representations. A quotient orientation may fail to define a *unique* root lift before child choices are supplied; this does not mean that a completed product lacks a root lift.

Conditional orientation signatures remain useful provenance and a structural validation oracle, but they must not enlarge the numerical evaluator domain. For every residue, verify that each completed signature is contained in the ordinary root domain and fail contextually otherwise. Numerical evaluation, threshold geometry, tropical sampling, public labels, and explicit orientation selection all continue to use the ordinary root orientations alone. Remove the staged runtime and standalone-schema plumbing that treated conditional signatures as supplemental evaluator sectors unless a future topology provides a concrete counterexample to the containment proof.

For the top-bubble acceptance failure, evaluator support is therefore not the current suspect: the measured child-only fits are

$$a_U = -2.0011044489, \quad a_H = -2.0011035828.$$

The implementation must next compare the separately retained (U), (S), and (US) terms on the identical numerical soft ray and locate where the required leading cancellation in (S-US) is lost. A symbolic assertion that (H-U) is nonzero is necessary but insufficient; the branchwise leading powers and coefficients must demonstrate the expected two-power improvement before the physical profile can pass.

The conditional factor tree is not required to reproduce the undeformed, selector-valued root CFF term by term, or even to define a separately proved raw CFF identity. Write the orientation-summed deformed root and conditional representations as (R(s)) and (C(s)). For the degree-two top bubble, the operator-level compatibility condition is precisely

$$J_s^{\leq 1} [R(s) - C(s)] = 0.$$

The two root-only sectors may therefore be absent from (C(s)) provided their constant and linear soft content is redistributed among the changed conditional residues. A difference beginning at order (s^2) is immaterial to (S₁). Tests must compare the completed, orientation-summed (S) and (US) jets with residues of their 4D-expanded counterparts; a raw pre-jet equality is neither a substitute nor an acceptance requirement.

Authoritative diagnostic: quotient-first rebuilding deletes the required soft jet

The branchwise massive-top-bubble profile locates the first numerical failure more precisely than the structural factor-tree tests. On one identical summed-orientation soft ray, the undeformed root and ordinary branch have the expected reported scaling near (-2), while the completed occurrence-aware soft jet *before* conditional rebuilding has the same leading coefficient with the opposite forest sign:

$$a_{\text{root}} = -2.00111449, \quad a_{S, \text{pre-rebuild}} = -2.00110944.$$

At the first and last sampled points, their imaginary parts combine as

$$\begin{aligned} 1.7181377 \cdot 10^{-7} - 1.7184258 \cdot 10^{-7} &= -2.88 \cdot 10^{-11}, \\ 1.7462234254 \cdot 10^8 - 1.7462234257 \cdot 10^8 &= -2.93 \cdot 10^{-2}. \end{aligned}$$

The remainder is two powers softer. Thus the occurrence-aware deformation, its physical-mass handling, its numerator route, and the (S₁) Taylor projection have already produced the low-order polynomial required by the physical acceptance test.

The final `rebuild_conditional` call changes that same completed branch to

$$a_{S, \text{conditional}} = +0.00135784,$$

and changes the completed (US) branch similarly to approximately zero soft scaling. The reconstructed branches are about two powers too soft to cancel the root and ordinary contributions. The failure therefore begins at the 30-orientation root-to-28-product reconstruction, not in CFF occurrence ownership, the canonical momentum route, the Taylor endpoint, or evaluator orientation support.

Concretely, the current conditional tree computes

$$\sum_{\alpha=1}^{14} Q_{\alpha}(q) J_s^{\leq 1} \sum_{\beta=\pm} B_{\beta|\alpha}(sq),$$

but implements no redistribution which would make it equal to

$$J_s^{\leq 1} \sum_{\omega=1}^{30} R_{\omega}(q, sq).$$

The two expressions agree only on the undeformed physical diagonal if the appropriate causal identities are used; that does not imply equality after the child occurrence of the boundary carrier is scaled independently of its co-graph occurrence. In particular, root orientations 0 and 29 have no conditional product, and the current 28 changed products do not inherit their constant or linear jet content.

The immediate implementation audit must therefore treat the completed full-root occurrence-aware jet as the reference production value. Before removing conditional rebuilding, re-run the DGSE reduced-parent and simultaneous-hard tests with both (S) and (US) retained on that same representation. If those limits fail, repair how a *containing* operator acts on the completed child polynomial; do not replace the child polynomial by the already-falsified quotient-first atom. The conditional tree may remain a topology/provenance diagnostic only if it is not used as a physical branch.

The same-component overlap must remain on the same representation as the soft branch. Preserving only (S) while rebuilding (US) still leaves the top-bubble fit at approximately (-2). Preserving both completed full-root branches makes the existing non-vacuous child-only acceptance test pass: the ordinary control retains raw behavior near (q_g^{-5}) , while the complete soft-refined result behaves near (q_g^{-3}) , i.e. the reported bulk scaling improves from approximately (-2) to approximately (0). This is also a direct numerical check of the ordered definition (US=U(SX)); (S) and its overlap may not be evaluated in different residue representations.

Containing-parent discriminator: a full-root child is necessary but not sufficient

Keeping both the child (S) branch and its same-component (US) overlap on the completed full-root CFF fixes the massive-top-bubble soft limit, but applying the same representation indiscriminately through every containing operation fails the DGSE hard limits. In the matched DGSE profile, the full-root result has

$$a_{\text{soft}} = +4.06730619, \quad R^2 = 0.99965276,$$

while five of 27 resolved UV limits fail. Each failed ray fixes the child edge (e5) and scales a containing or complementary loop direction; its fitted slope is (+2). The otherwise identical all-(U) control fails zero of 27 limits. This rules out the universal two-line bypass of conditional rebuilding: the full-root representation is the correct completed child soft polynomial, but it is not automatically a valid coordinate representation for every strict containing hard projection.

The next admissible implementation is therefore operation-boundary aware:

- retain the physical child (S) and same-component (US) values on the occurrence-aware, orientation-resolved root CFF;
- apply a strict containing parent (U) directly to each completed child term, using the containing component's homogeneous route and active-loop scope;
- retain the chronological deformation program so that a later operator sees the actual child polynomial and does not infer its centre from its final algebraic form;
- never reconstruct the child (S) polynomial from quotient-first factors, since the top-bubble result has already disproved that equality.

For each completed child history (B), any proposed conditional parent chart must satisfy, coefficient by coefficient,

$$J_u^{\leq d_r} \Phi_\Gamma^U \{ N_{\Gamma/\gamma} [C_{\text{cond}}(B) - R_{\text{root}}(B)] \} = 0,$$

without summing distinct orientations and with the same canonical route, mass provenance, selector, and tensor contraction. For the available top-bubble and DGSE strict parents, (d_-), so the constant coefficient alone is a sharp first discriminator. A passing hard profile is not proof of this identity; the branchwise recurrence and the matched profile must both pass.

Corrected representation verdict: retain the orientation-resolved CFF

The branchwise and hard-limit diagnostics rule out the current conditional reconstruction, but they do **not** imply that exact raised residues are a new requirement of the soft scheme:

- quotient-first conditional rebuilding deletes the constant and linear soft jet needed by the massive-top-bubble limit;
- retaining the completed root child jet restores that soft limit, while the current containing-parent route/scope then violates strict hard projections in DGSE;
- combining a full-root child with a conditional strict parent fails both the exact outer-pair recurrence and fourteen of twenty-seven DGSE UV rays;
- discarding the strict degree-zero parent image of the refinement, although its exact four-dimensional value is zero, still fails the same fourteen three-dimensional rays.

The exact four-dimensional DGSE topology independently establishes

$$U_{\Gamma,0} [(H_\gamma - U_\gamma) C_{\Gamma/\gamma}] = 0.$$

Ordinary UV counterterms already generate raised four-dimensional propagator powers. The established 3D implementation does not construct their residues explicitly. Instead it generates the ordinary CFF per orientation, substitutes the corresponding on-shell energies in the numerator, applies the complete UV deformation to that moving 3D expression, and only then extracts its Laurent jet. Derivatives of pole locations, numerator energies, causal surfaces, and the 3D measure are therefore supplied by the chain rule within each orientation. Raised propagator powers alone cannot explain a failure first seen for soft counterterms.

The genuinely new requirement is the ordered composition of deformations with different centres and grading rules. For every orientation (ρ), the production path must implement

$$H_{\delta,\rho}C_\rho = U_{\delta,\rho}C_\rho + S_{\delta,\rho}C_\rho - U_{\delta,\rho}(S_{\delta,\rho}C_\rho),$$

and a containing operator must act on each complete signed child term on that same orientation-resolved representation. The orientation selector itself is fixed; only the component-owned momenta, energy shifts, masses, and surfaces follow the selected deformation. No counterterm may rely on cancellation between different orientations to become locally subtracted.

The failed full-root experiment therefore diagnoses an incorrect containing route or active-variable scope in the retained replay, not missing raised-residue data. The failed conditional experiment diagnoses a change of mathematical CFF representation under an exceptional component-local deformation. Repair those operation scopes on the established full CFF path; do not convert the result to quotient-first denominators after the child jet has been formed.

An independent positive-energy residue oracle with a linear energy numerator now tests this mechanism explicitly. For

$$D = k_0^2 - E^2, \quad A = 2(k_0 p_0 - \ell p_1), \quad N_0 = a + b k_0, \quad N_1 = c + d k_0,$$

the first two coefficients of

$$\frac{N_0 + s N_1}{D^2 + s A D}$$

at the positive pole give

$$R_0 = -\frac{a}{4E^3}, \quad R_1 = -\frac{c}{4E^3} + \frac{p_0 b}{8E^3} + \frac{3\ell p_1 a}{8E^5}.$$

Multiplication by a strict parent's positive-pole factor yields

$$P_0 = -\frac{a}{8FE^3}, \quad P_1 = -\frac{c}{8FE^3} + \frac{b}{16E^3} + \frac{3\ell p_1 a}{16FE^5}.$$

The terms proportional to derivatives of the numerator and cograph remain a useful oracle for a future implementation which derives local 3D counterterms from expanded 4D quantities. They are not a Phase-1 production gate and do not justify replacing the per-orientation CFF construction.

The current conditional factor tree remains useful as topology and ownership provenance, but it is not a valid production replacement for a nested soft atom. Phase 1 continues on the established orientation-resolved CFF path.

Canonical spatial-momentum representation

There is one persistent abstract spatial-vector convention:

$$Q3(e, \text{mink}(4)) \quad \text{or} \quad Q3(e, \text{mink}(4, \mu)).$$

The edge argument is a non-negative integer graph-edge ID. A bare $Q3(e)$ is invalid. Explicit components are stored as $Q(e, \text{cind}(i))$, not as a second $Q3$ convention. Internally projecting $Q3$ onto $\text{cind}(\theta)$ normalizes to zero, while projecting onto a spatial $\text{cind}(i)$ normalizes to $Q(e, \text{cind}(i))$. Malformed edge arguments must remain visible long enough for graph validation and may not be normalized into Q first.

Rejected investigation: summed exact 4D-to-3D bridge

Do not implement this bridge. The following records the investigated approach so that its failure mode remains explicit. It was rejected because a summed residue hosted on one orientation does not subtract any individual orientation and therefore cannot be used for local sampling, profiling, contour deformation, or residue selection.

The experiment was rejected: an orientation-summed residue cannot serve as an orientation-local counterterm. Hosting it on one selector only prevents duplicate counting; it does not subtract the other orientation channels. Consequently it is unsuitable for local sampling, profiles, contour deformation, selected cuts, or residue-specific evaluation. No part of this bridge is connected to production.

Corrected Phase-1 implementation boundary

Reuse the existing per-orientation CFF representation and extend its retained deformation history so that:

- (S) deforms only occurrences owned by its connected component and preserves physical masses;
- (US) applies a fresh ordinary deformation to the complete (S) polynomial on the same orientation;
- a parent applies its selected deformation to every complete signed child term, with the child boundary carrier reclassified according to the parent loop system;
- selectors and unowned co-graph occurrences stay fixed;
- normalization combines orientations only after each one has received its complete forest subtraction.

Because the counterterm is constructed directly on the CFF, cancellation is an orientation-local requirement, not merely a property of the summed orientation result. UV profiling must therefore be run for each orientation individually, using generation-time orientation selection where useful to shorten the diagnostic loop. A failure report must retain the orientation signature/selector together with the routed hard ray so that the faulty surface ownership or expansion route can be isolated. The summed-orientation profile is an additional end-to-end check and may not replace these per-orientation checks.

The positive-energy raised-pole test remains an independent algebraic oracle for future 4D-derived counterterms. It is not wired into Phase-1 production.

This branch does not yet contain the separate work supporting original numerators with more-than-linear dependence on an individual EMR energy. Phase-1 therefore selects fixtures whose original numerator is already at most linear in each EMR energy (in particular, quark-loop examples are valid, whereas the gluon-loop contribution to a gluon self-energy is excluded). This is a temporary fixture precondition only: do not add a degree analyser, an early rejection, or any other production machinery that would immediately be removed when that branch is merged.

Authoritative mixed-surface and orientation verdict

This section supersedes every earlier research-log passage proposing partial mixed-surface deformation, conditional-CFF regeneration, supplemental orientations, summed-only validation, or a quotient-only containing operator. Those passages are retained only as the record of rejected investigations.

The connected soft operator owns a CFF denominator occurrence only when both its stored topology shore and its energy/shift support are local to the active connected component (including its crown carriers). Such an occurrence receives the complete component-local spatial and temporal deformation. A pure co-graph or mixed/spanning parent-CFF shore receives no deformation and is restored literally from its occurrence tag. There is no production `Partial` ownership class.

The rejected partial rule mapped a mixed denominator schematically as

$$\sum E_{\text{child}} + \sum E_{\text{cograph}} + q_0 \mapsto \text{prepare}(\sum E_{\text{child}}) + s(\sum E_{\text{cograph}} + q_0).$$

At $s = 0$ this erased the co-graph causal suppression. In the selected DGSE orientation every causal denominator was mixed, so the pure child atom lost all quotient causal denominators and five quotient-hard rays incorrectly scaled as $+2$. This was not evidence for a quotient-only outer operator. Production keeps the full containing `FullSubgraph` ordinary operator required by the forest recurrence.

CFF generation records each concrete occurrence's generating shore and support before algebraic cache deduplication. The occurrence lift freezes every tagged inverse denominator before applying component-local substitutions. Therefore an unowned occurrence remains exact even when its mathematical denominator equals an owned occurrence; only the owned occurrence receives the transformed image. Ambiguous or missing provenance must still produce a contextual error rather than a global replacement.

The direct 3D validation for this correction is orientation-local. For DGSE, generation-time filtering of orientation $(-, -, 0, 0, +, +, +, 0, +)$ leaves exactly $--00+++0+$; all 27 routed UV limits then pass individually, including the five former $+2$ failures. Unfiltered profiling covers 30 orientations times 27 limits. Some individual fits have pre-asymptotic zero crossings in the original 10^8 -to- 10^{12} window; over a 33-point 10^8 -to- 10^{16} window all 810 resolve below -0.9 , while the isolated problematic channels converge to -1 over 10^{12} -to- 10^{16} . Acceptance must retain orientation labels and hard rays, and must choose a window which actually reaches the orientation's asymptotic regime rather than classifying a zero crossing as a subtraction failure. Summed profiling remains an additional check only.

A pre-residue 4D deformation can give a different image for a selected mixed pole. That fact is a negative algebraic oracle showing that operation-before-residue does not define direct CFF ownership; it must not be connected to the Phase-1 production operator. Exact 3D-versus-4D local evaluation remains a later merge gate once local counterterms derived from expanded 4D quantities are available.

Tier-1 cleanup of the rejected conditional replay

The quotient-first conditional factor tree and its deferred replay program were an experimental implementation of the representation change rejected above. They are not a dormant alternative production path: rebuilding a completed soft jet on those factors deletes the constant and linear content needed for the top-bubble cancellation and changes the evaluator domain. The module, replay source, replay scopes, factor-local deformation callback, completed-factor direct-map branch, and tests whose only oracle was that representation are therefore removed.

The retained jet history records only the connected component, operator kind, Taylor parameter, and diagnostic label needed by the active full-CFF forest recurrence. A containing operator continues to act directly on every completed signed child term in the same orientation-resolved CFF. The serialized `supplemental_orientations` field and the broader CFF-generation APIs remain in place for schema compatibility and possible future topology work; Phase 1 does not populate them through conditional replay.

Current direct-CFF refinement: contraction-transition ownership

The later orientation-local audit supersedes the preceding claim that a mixed parent-CFF occurrence must always remain literal. That rule preserves the hard behavior but omits the child soft jet. Conversely, deforming every mixed factor makes all four DGSE denominators coalesce at the exceptional point and removes three quotient suppressions. The missing distinction is not the cached surface, shore containment, or edge support alone; it is the role of a factor in one concrete CFF contraction branch.

For a root-to-leaf CFF contraction history with current vertex partition Π , define

$$b_\delta(\Pi) = \#\{B \in \Pi \mid B \cap V_\delta \neq \emptyset\}.$$

A transition which merges blocks A and B belongs to the component contraction tree exactly when

$$A \cap V_\delta \neq \emptyset, \quad B \cap V_\delta \neq \emptyset.$$

It then decreases b_δ by one. Restricting the full contraction dendrogram to V_δ , deleting outside leaves, and suppressing unary merges leaves precisely these transitions. Hence every path contains $b_\delta(\Pi_{\text{seed}}) - 1$ component factors, or $|V_\delta| - 1$ for the singleton seed partition of the root CFF. This is topology neutral and also states the correct generalization when a future factor starts from precontracted blocks.

For DGSE and $\delta = \{v_2, v_3; e_5, e_6\}$, the selected orientation `--00+++0+` has the chain

```
{v2}+{v1}
{v1, v2}+{v5}
{v3}+{v4}
{v3, v4}+{v1, v2, v5}
```

Only the terminal transition joins two blocks which both meet the child. Its mixed denominator is the component-reducing causal factor and receives the complete deformation

$$D_i(s) = I_i(k + sq) + sB_i(q),$$

where I_i is its component-internal energy current and B_i its complementary boundary flux. This occurrence is classified as `Complete`. The other three mixed denominators are spanning/quotient occurrences. When one contains a component-internal energy, that internal dependence still receives $I_i(k + sq)$, because the parent CFF is not factorized into separate component and co-graph CFFs, while its complementary boundary flux $B_i(q)$ remains literal. Such an occurrence is classified as `InternalOnly`. An occurrence containing no component-internal energy remains entirely literal and is classified as `None`. Across all 30 acyclic DGSE orientations and 70 root-to-leaf monomials there is exactly one `Complete` component factor per monomial. Shore restriction alone is ambiguous, and 24 shared CFF nodes have different roles on different outgoing branches.

The minimal sufficient provenance is therefore the selected block and its actual adjacent contraction partner on each parent-to-child transition, plus the remaining partner at a terminal leaf. A node-only occurrence tag is insufficient because its inverse denominator is factored outside the child sum. The direct-CFF representation must distribute that factor over outgoing branches and give each transition a distinct occurrence identity before Symbolica collapse. Equal mathematical denominators remain equal after the unit-valued transition tags are stripped.

The active implementation records only this topology-neutral transition metadata during the existing CFF generation, branch-expands the occurrence-tagged form, applies the three deformation modes above, and keeps the ordinary parent operator as `FullSubgraph`. The focused orientation's 27 hard rays, all 810 unfiltered orientation/ray pairs, and the matched soft profile pass. Failures must still be diagnosed on the same orientation before any summed result is considered.

Nested-jet representation audit

The contraction-transition rule solves *which* causal factor a soft operation owns, but a unit-valued tag on a flattened inverse is not by itself sufficient to compose differently centred completed jets. After an ordinary projection,

$$J_u[f(u)g(u)]$$

contains product-rule cross terms. The tag originally attached to f^{-1} has factorized away from its coefficient, so replacing that tag cannot freeze or transform the exact completed occurrence independently of g . In particular, using

the soft `Complete/None` classification as a record of which factors a previous ordinary U changed is invalid: U acts on every factor carrying the active loop variables, whereas S owns only the component-reducing contraction transitions. Silently restoring the cached pre-jet denominator would corrupt nested U/H terms.

The mathematically clean direct-CFF representation is therefore to retain an opaque occurrence factor

$$\mathcal{D}_\omega[D_\omega] \leftrightarrow D_\omega^{-1}$$

for each orientation/branch occurrence ω until every nested deformation has been composed. Ordinary U transforms the stored argument D_ω wherever its variables require it. Soft S transforms that argument only for component-reducing transitions and holds every other occurrence wrapper fixed while applying the component map to the numerator, tree factors, and other exposed structures.

Each retained forest term then stores the fully composed multivariate deformation together with its ordered jet list, but does not project a jet during composition. At materialization the occurrence wrappers are lowered back to inverse causal denominators and the jets are evaluated in their original order, from the innermost child outwards. For two stages this uses

$$J_s \Phi^S(s) J_u \Phi^U(u) X = J_s J_u \Phi^S(s) \Phi^U(u) X,$$

where the second equality only moves coefficient extraction past a later substitution which is independent of u ; it does not exchange the two deformation maps or assume $US = SU$. Thus all product-rule cross terms are generated by the final multivariate expression, while the occurrence identity is still available when each deformation is composed. This remains a direct, per-orientation operation on the parent CFF and introduces no 4D reconstruction.

Until this representation is implemented, a completed prior jet followed by a soft operation which must leave one of its transformed causal occurrences literal must fail closed rather than use the cached denominator as a surrogate.

Authoritative implementation-state addendum: immutable edge scope, materialization boundary, and canonical routes

For the direct 3D CFF operator, ownership of every exposed noncausal inverse-energy or tree-propagator factor is determined only by its immutable physical graph-edge ID. In `LOCAL_3D_OSE_EDGE(e_phys,e_basis,mass_kind,D)`, `Den(e_phys,...)`, and `raw OSE(e_phys)`, `e_phys` is the ownership key; the mutable expansion-basis edge is never an ownership key. Before a connected deformation, freeze occurrence-wrapped causal inverses first and then freeze every remaining edge-local noncausal factor whose physical edge is outside the active component. Route preparation and the selected deformation act only on the exposed owned factors. Restore each frozen factor literally before Taylor projection, so it is constant in that stage but still participates in the complete product. The same factor is exposed when a containing component owns its physical edge. A prepared `OSE(basis,D)` without an immutable physical-edge ID, a malformed or out-of-range ID, or a private freeze wrapper surviving to materialization is a contextual error.

Materialization is a one-way boundary for occurrence-local composition. It lowers the exact-current occurrence wrappers and projects the stored jets; the resulting coefficients contain numerator/denominator product-rule cross terms which cannot in general be refactorized into the original owned occurrences. Unit-valued occurrence tags or cached pre-jet denominators do not restore that lost factorization. Consequently, applying a fresh eager $\top$ mapper to a materialized atom globally reinterprets the completed CFF and is not an oracle for the orientation-local action of a containing operator. Production composes every later deformation on the retained deferred term before materialization and projects the jets in chronological child-to-parent order. Structural tests inspect that retained history; hard-limit acceptance is supplied by the real per-orientation `profile ultra-violet` path, with the summed profile as an additional check.

A forest node's nominal `current.lmb()` is enumeration-path metadata, not the semantic chart of its counterterm. The same connected component may otherwise appear with equivalent representatives such as `[e2,e3]` and `[e2,e4]`. Both 3D and 4D operations therefore first use the graph-canonical full route or the compatible component subroute induced from it, and every accepted loop coordinate must be homogeneous with respect to graph-external momenta. The 3D full chart is derived deterministically from that component chart and the operation's genuinely independent boundary carriers; an incoming child chart is validated and converted but does not choose a different parent chart. In 4D, the completed child records the coordinates on which either its selected atom or its all-ordinary reference actually depends. Physical membership of an otherwise absent coordinate representative in the contracted child is not an additional chart constraint. If the canonical candidate cannot retain a recorded coordinate, enumerate only compatible homogeneous bases and choose the edge-lexicographic minimum. This deterministic fallback is permitted only for such genuine retained coordinates. Never select the nominal forest LMB or a traversal-dependent first compatible basis, and reject any required affine external-momentum shift contextually.

Authoritative implementation correction: reuse the orientation-local MUV map

The occurrence/shore and deferred-multijet designs above are superseded by a direct audit of the existing, working 3D MUV pipeline. The ordinary operator does not identify or replace individual E/H-surface occurrences. It starts from the complete orientation-resolved parent CFF, rewrites edge momenta into the active connected component's compatible local LMB, and applies its deformation to that expression as a whole. The soft operator must use the same scope and route; only its deformation differs.

For every edge momentum resolved in the operation LMB as

$$Q_e = L_e + P_e,$$

ordinary large-loop scaling uses the existing representation $L_e/t + P_e$, whereas soft scaling uses $L_e + s P_e$. The latter is applied to both $Q3(e, \text{mink}(\dots))$ and explicit $Q(e, \text{cind}(\dots))$ occurrences, including the temporal external-energy shifts already present in the parent CFF. Loop variables, on-shell energies, and physical masses remain fixed in the soft deformation, and no UV mass is introduced. The Taylor parameter is projected before the completed child atom is returned, so a parent applies its fresh graph/LMB deformation to that actual completed expression just as it already does for nested ordinary counterterms.

Accordingly, production must not attach topology-shore identities, freeze individual causal denominators, retain deferred jets, generate component CFFs, or reconstruct a 3D counterterm from a 4D one. Equal mathematical CFF factors remain equal. Component locality is supplied by the graph subgraph and the compatible local LMB replacements already used by MUV. Validation is per-orientation structural inspection and per-orientation UV/soft profiling; summed-orientation checks are additional rather than substitutes.

Superseded implementation-state addendum: one parent-CFF convention

The temporary normalized component/cograph CFF-block builders are not part of the direct 3D counterterm design and have been removed. Their proposed per-block orientation inventory, residue-count transport, and generated-occurrence namespace encoded the rejected factorized replay approach described in the superseded findings above. No occurrence-orientation indexing convention remains in the exporter: it consumes the already-selected graph-term orientations directly, and no generated-block namespace or second orientation filter exists.

`generate_uv_cff` remains an unnormalized fixed-orientation graph diagnostic. It may demonstrate, for example, that contraction turns a root-acyclic orientation into a cyclic cograph, but local soft counterterms never use its result as a component atom or multiply it into the parent expression. Production starts from `Graph::cff` or `Graph::cff_for_orientations`, attaches occurrence identity to that parent CFF, and composes every U/S deformation directly on its retained orientation-resolved terms. Contracted undirected edges receive a temporary direction only inside the diagnostic builder before the edge is removed; this is not a second orientation or momentum convention.

Final direct-CFF rule: stage-relative pole energies

The existing MUV inspection trace fixes the precise meaning of the preceding direct-CFF rule. For a connected component δ and a completed orientation atom, first route only the reduced component $\delta \setminus \gamma$ through its compatible local LMB. If

$$Q_e = L_e + P_e,$$

then the soft deformation is applied to the exposed component dependence as

$$L_e + P_e \mapsto L_e + s P_e.$$

The scaled generators are $Q3(e, \text{mink}(\dots))$ and explicit $Q(e, \text{cind}(\dots))$ for physical edges in `dummy_less_full_crown(delta)`. Although a compatible sub-LMB also retains the original graph-external generators, those are spectators unless their physical edges belong to this component crown. The crown is that of the complete current component, not the reduced graph (whose holes are internal child boundaries) and not an active disconnected-sector union.

A remaining one-argument $05E(e)$ is held literal during this stage. It is a pole energy of the already retained parent residue, not an independent component-external Taylor generator. Scaling it globally would also deform the universal parent-CFF factors and co-graph causal denominators. For the massive-top bubble this changes the intended Laurent form from $s^{-2}F(s)$ to $s^{-4}F'(s)$, so the fixed endpoint -1 no longer selects the component's $j = 0, 1$ Taylor coefficients.

In contrast, an active internal energy has already been materialized by `start` as `0SE(lmb_id,D[Q3])`; its argument is transformed through the routed `Q3` dependence.

This distinction is stage-relative rather than permanent ownership metadata. An `0SE(e)` held fixed by a child becomes an internal energy when a containing component owns edge e . The containing operation's fresh `start` then materializes it as `0SE(lmb_id,D[Q3])`, after which the containing deformation acts on its argument. This is how eager child completion composes with a parent operation without surface-occurrence IDs or deferred jets.

The direct 3D prescription is therefore:

1. retain the complete parent CFF separately for every orientation;
2. materialize the active internal energies with the existing `start` step;
3. route the reduced component with the same compatible local LMB machinery as MUV;
4. scale only the current component-crown `Q3/Q` generators, keeping raw parent-residue `0SE`, masses, selectors, and graph-external spectators literal;
5. multiply by s^{-d} , keep Laurent powers through -1 , and set $s = 1$;
6. apply the existing ordinary 3D MUV operator to this completed soft atom for the US branch, then combine $H = U + S - US$.

This is an orientation-local CFF jet at fixed retained pole energies. It is not obtained by taking a residue of the implemented 4D soft counterterm, and Phase 1 does not claim per-term equality to such an operation. Its current acceptance oracle is cancellation in every orientation: in the top-bubble fixture the eight problematic ordinary-U orientations scale near -2 , while all corresponding H orientations scale near zero and improve by two powers; a containing degree-zero U changes the result only at numerical-noise level. The nested H/H fixture has 315 resolved orientation-level UV fits with no failure, while its matched bare control fails 183.

Finally, the local-only forest algebra does not admit a positive-degree ordinary-U parent above a positive-degree H child in general. The finite $(1 - U)S$ child refinement needs the containing soft completion appearing in the paper's nested formulas; a degree-zero parent is harmless because $H_0 = U_0$. Until the deferred scheme-change/integrated policy exists, production therefore accepts U/U, U/H, H/H, and H/U only for a degree-zero U parent, and rejects positive-degree H-child/U-parent woods contextually.

Authoritative correction: hard-dual soft deformation in the parent CFF

The preceding stage-relative crown-scaling rule is superseded by the hard-dual construction below. Its restriction on a positive-degree ordinary parent is independent of that representation choice and remains necessary, as shown by the mixed-wood counterexample at the end of this addendum. Ordinary UV Taylor expansion already admits two exactly equivalent homogeneous charts: scale external momenta and masses to zero, or scale the component-internal loops to infinity. The soft Taylor expansion has the same freedom, with a different mass grading.

Let a connected component integrand without its loop measure have homogeneous engineering degree h , with internal loop momenta k , component-external momenta p , and all dimension-one scales m and any existing terminal UV scale M held fixed by the paper's soft limit. Then

$$X(k, sp, m, M) = s^h X(k/s, p, m/s, M/s).$$

Consequently the two hard representations are

direct small-variable chart	dual hard chart
$U(p, m) \mapsto (sp, sm), M \text{ fixed}$	$(k, M) \mapsto (k/s, M/s),$
$Sp \mapsto sp, (m, M) \text{ fixed}$	$(k, m, M) \mapsto (k/s, m/s, M/s).$

The hard chart does not change the definition of S , expand a physical mass, or introduce M_{UV} . Co-scaling m is precisely the homogeneous representation of leaving that physical mass fixed in $X(k, sp, m)$.

For an edge routed in the component LMB as

$$q_e = L_e + P_e,$$

the soft hard deformation is

$$q_e \mapsto P_e + \frac{q_e - P_e}{s} = P_e + \frac{L_e}{s}, \quad m_e \mapsto \frac{m_e}{s}.$$

Thus it is a dilation of the internal part around the fixed external-flow point P_e , not an affine change of loop basis. In the 3D representation,

$$E_{m/s}(L/s + P) = s^{-1} E_m(L + sP).$$

An E- or H-surface containing active energies therefore generates the correct soft scaling of its spatial dependence and temporal boundary shift automatically. No component crown, raw boundary OSE, tree denominator, CFF surface, shore, or contraction occurrence is globally rewritten. Mixed surfaces are handled in exactly the same orientation-local manner as by the existing MUV hard-loop map: active energies are deformed and spectator energies remain fixed.

With L active residue loops and 3D homogeneous degree h_{3d} ,

$$s^{-3L} X(k/s, p, m/s) = s^{-d} X(k, sp, m), \quad d = 3L + h_{3d}.$$

If the direct soft expansion is $\sum_{j \geq 0} X_j s^j$, its hard form is $\sum_{j \geq 0} X_j s^{j-d}$. The fixed Laurent endpoint -1 therefore selects exactly $j = 0, \dots, d-1$, while $d = 0$ has no soft branch. In the code's pre-inversion variable $x = 1/s$, scale active loop generators and component-local fixed mass scales by x , apply the existing x^{3L} measure grading, replace x by $1/s$, and retain strictly negative Laurent powers. There is no additional manual factor s^{-d} .

The active OSE normalization before inversion is

$$\text{OSE}(e, D) \mapsto x^2 \text{OSE}\left(e, \frac{D}{x^2}\right)$$

inside the outer square root. This extracts the expected hard energy factor and keeps the OSE argument regular for Symbolica. It differs from ordinary MUV only in the finite argument: MUV retains its

$$x^2 \text{OSE}\left(e, \frac{M^2 x^2 + D - M^2}{x^2}\right)$$

rearrangement, whereas soft introduces no M .

Loop and OSE edge identities already provide all required CFF locality. The only information lost by an eager Taylor projection is the support of a free mass factor. For example, a fermion numerator may leave a standalone m , and an inner ordinary projection may leave a coefficient proportional to $m^2 - M^2$. Two disconnected components can use the same model mass symbol, so a global mass substitution would act on the wrong sibling. The minimal persistent metadata is therefore a unit-valued mass-support tag anchored to a representative physical graph edge of the connected component. It is attached only to masses in the newly introduced component numerator and materialized active OSEs. Ordinary U-generated M receives the same support tag. A containing operation rebases contained tags to its representative; a disjoint operation leaves them unchanged. Soft scales only its active tag. The tag is replaced by one only on the cloned final-evaluator expression, after all forest operations have composed.

This mass-support tag is not CFF occurrence provenance. Equal causal denominators remain mathematically equal, and there are no shore IDs, surface ownership tables, deferred multijets, component CFF products, or 4D-to-3D counterterm reconstruction.

For disconnected components with

$$I = T(p)A(k_A, p, m)B(k_B, p, m),$$

the two hard maps use independent loop supports and independently tagged mass occurrences. Hence

$$S_B S_A I = T(p)(a_0 + a_1 p)(b_0 + b_1 p),$$

including the $a_1 b_1 p^2$ term. The shared bridge $T(p)$ is fixed. This is precisely the term lost by one global total-degree crown Taylor series. For nested components, the child projection is completed first; the parent then scales every loop and mass support contained in the parent. The maps need not commute, and their eager chronological order is exactly the forest product $K_{\text{parent}} K_{\text{child}}$. This establishes the implementation order, but it does not by itself make every mixed operator assignment locally finite. For $D_\gamma = H_\gamma - U_\gamma$, the change from a U/U forest to an H-child/U-parent forest is

$$R_{H/U} - R_{U/U} = -(1 - U_\Gamma) D_\gamma C_{\Gamma/\gamma}.$$

A child boundary monomial of degree $j \leq d_\gamma - 1$ has reduced-parent degree $d_\Gamma - d_\gamma + j$, which can reach $d_\Gamma - 1$. Thus a positive-degree ordinary parent can leave a nonnegative reduced-parent hard limit even though D_γ is finite in both the child-hard and simultaneous-hard limits. The existing exact algebraic and graph-level counterexamples verify this obstruction. Production therefore continues to allow a degree-zero U parent above H, but rejects a positive-degree U parent until either the ancestor is also assigned H or an explicit finite scheme-change/effective-vertex counterterm is implemented. This is separate from, and in addition to, the ban on integrated soft/PolePart mixtures.

The active implementation follows this rule directly on every parent-CFF orientation. Its first validation set establishes:

- homogeneous $d = 0, 1, 2$ hard/soft micro-oracles, including a materialized OSE and a fixed tree/co-graph denominator;
- a nested child whose spatial and temporal boundary dependence is softened while graph-external and raw parent-pole factors remain fixed;

- the exact two-component degree-two scalar spectacles fixture, including its summed and per-orientation UV profile and matched bare/MUV controls;
- consecutive soft components;
- a nested massive H/H top self-energy with a non-vacuous per-orientation UV profile; and
- all 30 orientations of the massive-top bubble child, including the eight ordinary-U orientations with the expected approximately two-power soft improvement under H.

These checks derive S from the 3D CFF itself. The 4D identity above is an algebraic proof and test oracle only; it is never used to construct or map the 3D local counterterm.

Verified mixed-wood and terminal-mass correction

The degree-qualified mixed-wood restriction above has now been tested by temporarily enabling all four production rows (U/U), (H/U), (U/H), and (H/H) in the same two-loop scalar fixture, retaining the four forest families separately and measuring the child, reduced-parent, and simultaneous hard rays. Exactly one obstruction occurs:

$$H/U : \quad -(1 - U_\Gamma)H_\gamma I = O(\lambda^0)$$

in the reduced-parent ray, and the complete (H/U) forest has the same logarithmic remainder. The other three rows are locally improved in every required ray. This is direct production algebra, not a 4D-to-3D comparison, and confirms that the original unqualified request to allow a soft child below an ordinary positive-degree parent cannot be part of the local-only Phase-1 acceptance. A logarithmic parent remains allowed because (H_0=U_0); this is the mixed nesting exercised by the top-bubble vertex.

The same audit corrected an earlier malformed terminal-mass oracle. GammaLoop uses `mUVexp` as denominator provenance and `mUV` as the actual vacuum mass:

$$\text{den } (e, q, m_{UV, \text{exp}}^2, q^2 - m_{UV}^2).$$

An outer inverse-hard (U) keeps a free numerator `mUVexp` fixed, which gives it the required direct-Taylor mass grading, while it sends the actual terminal `mUV` to `mUV/s`. The terminal propagator then reproduces the same ($q^2 - m_{UV}^2$) coefficient and is not infrared-rearranged twice. The active regression now obtains this structure from a genuine inner `t_raw` projection before applying the outer operation; it no longer constructs an unphysical denominator with `mUVexp` as its actual mass.

Exact factorized materialization of the local soft operator

The separate $U(X)$, $S(X)$, and $U(S(X))$ expressions remain the definition and the small-oracle provenance of the local operator. Production 3D materialization uses the exactly equivalent linear form

$$H(X) = S(X) + U(X - S(X)).$$

This does not exchange the order of U and S , and assumes neither commutativity nor idempotence. The existing ordinary 3D projector is linear for a fixed component, active sector, and compatible LMB: its routing and mass scope substitutions are fixed ring homomorphisms, and multiplication by the measure factor, Laurent truncation, and final parameter evaluation are linear. Consequently,

$$U(X - S(X)) = U(X) - U(S(X))$$

with exactly the same component route and mass support as the two separately materialized branches. A parent still receives the completed child atom and applies its own factorized operation afterwards. HedgePoset forest nodes and their signs remain separate; only the internal representation of one completed H atom is changed.

This factorization supersedes earlier implementation wording which required production to retain two complete, nearly cancelling $U(X)$ and $U(S(X))$ atoms simultaneously. Focused algebraic and inspect tests must still compare the conceptual $U + S - US$ branches independently on expressions small enough to materialize them safely.

The immediate engineering reason is also mathematically aligned with the subtraction: on the one-orientation Figure-B.1 fixture, separately retaining the two UV branches produced a contracted scalar Add of 5,384,386,060 bytes. Symbolica stores Add lengths as 64-bit values but stores a Mul/Fun child length as 32 bits, so wrapping that scalar truncated the size by exactly 2^{32} and a later scalar-alias traversal panicked. Applying U once to the soft remainder allows equal Laurent coefficients to cancel during the series projection rather than after tensor contraction. Compact byte-size logging is retained at the local projection, forest aggregation, final-numerator, and evaluator boundaries. If a future graph still reaches the representation ceiling, the next clean boundary is to keep signed forest-node atoms as separate evaluator inputs and sum their numerical values per orientation; it is not to reconstruct the 3D counterterm from a 4D expression or to hide large branches in tagged function-map bodies.

Signed forest-node evaluator decomposition

The factorized local operator reduced, but did not remove, the representation ceiling in the complete one-orientation Figure-B.1 fixture. Its second contracted scalar was 5,241,104,034 bytes; the enclosing Symbolica child length again retained only 946,136,738 bytes, differing by exactly (2^{32}) . The failure occurred after the direct-CFF forest had been completed, in tensor network scalar reconstruction. It is therefore neither evidence for a different soft deformation nor permission to reconstruct the counterterm from the four-dimensional representation.

HedgePoset now retains every signed final node contribution as a named parametric evaluator term before its residue-wise analytical sum. The latter remains available only for consumers which explicitly require one symbolic expression, including legacy comparison and analytical inspection. Amplitude evaluator construction consumes the named terms directly. The existing multi-output evaluator machinery contracts and optimizes each term independently; after the requested orientation or orientation sum has been evaluated, the amplitude wrapper adds every returned value. No output may be dropped, and an empty output list is an error.

This changes only the association of the exact forest sum. It does not split one connected approximation into independently marked (U), (S), and (US) counterterms: a completed (H) node remains one signed evaluator channel. It also retains the node key needed to diagnose any channel which remains too large. If one node alone exceeds the representation ceiling, that node may be partitioned further only along its existing top-level additive terms while retaining the same node identity; products are never split and no tagged function-map surrogate is introduced.

Because exact symbolic cancellation between different forest nodes is then performed numerically, acceptance requires the existing stability escalation and matched per-orientation UV profiles. The Figure-B.1 regression must both complete evaluator construction and pass its per-orientation profile; merely avoiding the 32-bit panic is not sufficient.

Superseding implementation-state addendum: forest-wide run-time aggregation

This section supersedes the proposed runtime behavior in **Signed forest-node evaluator decomposition** above, while preserving that section as the record of the investigated and rejected design. Signed forest nodes must remain separately available through the computed inspect/export path, with their node keys, parent keys, residue indices, local projection paths, and 4D branch provenance. They are not independently evaluator-ready runtime channels.

The evaluator-facing representation is the historical residue-wise sum of the complete signed forest. HedgePoset first applies per-node normalization and color collection, adds all compatible signed nodes, and only then performs the final orientation-local four-vector split. This ordering is required because temporal components in localized integrated-counterterm polynomials can cancel between forest nodes before an internal four-momentum is replaced by its orientation-resolved energy and spatial parts. Moving that split across the forest sum exposed spurious internal energy parameters and failed two sunrise subgraph profiles by two powers.

Consequently, `ParametricIntegrands` again contains one `Integrands` residue map, `AmplitudeDerivedData` again contains one completed `all_mighty_integrand`, and evaluator construction receives exactly that one atom. Standalone archives therefore retain their version-5 one-output layout; there is no additive-term count, logical-result post-reduction, or pairwise forest reduction. The rejected multi-output design first summed each term's contiguous orientation group and then used an adjacent-pair tree across terms; that association was deterministic, but it could not reproduce symbolic cross-node cancellation and therefore is not a valid replacement for the forest-wide sum.

This runtime aggregation does not weaken the requirement to test forest terms one by one. Small algebraic oracles and computed UV-forest exports continue to retain the four signed nested families independently before any test regroups them. Production evaluation alone consumes their completed symbolic sum.

Canonical 3D route transitivity audit

No additional retained-carrier or CFF-occurrence metadata is required for nested local-LMB replacement. Let B be the ordered cotree-edge set of the graph-canonical full LMB and let $T = G \setminus B$ be its spanning tree. For any (possibly disconnected) subgraph A , `try_compatible_sub_lmb` chooses the first $r(A)$ -element subset $R_A \subseteq B \cap A$ whose deletion leaves every component of A connected. Equivalently, start from $T \cap A$, scan the edges of $B \cap A$ in reverse canonical order, and add an edge exactly when it joins two current components. The rejected edges are R_A .

For nested subgraphs $C \subseteq P$, connectivity in the working forest for P is, at every point of that reverse scan, a coarsening of connectivity in the working forest for C . Therefore, whenever an edge is rejected from the child forest because its endpoints are already connected, its endpoints are already connected in the parent forest as well. Hence

$$R_C \subseteq R_P.$$

The argument applies componentwise and includes self-loops. Since `Spinney::with_scheme` induces every production component LMB from the same graph-canonical full LMB, every Q3/Q carrier retained by a completed child is necessarily a loop carrier of a containing parent. The parent hard map thus scales it through the existing MSbar-style local-LMB replacement. LMB generator ordering may differ, but the physical carrier-edge set used by the replacement is nested, which is the required invariant. This rules out the suspected frozen-child-carrier failure without introducing shore IDs, occurrence IDs, or a 4D reconstruction path.

Final acceptance reconciliation

The occurrence-ownership and ambiguity-rejection bullets in the original plan, including the corresponding completion bullet, are superseded by **Authoritative implementation correction: reuse the orientation-local MUV map** and **Authoritative correction: hard-dual soft deformation in the parent CFF**. They are retained above only as the investigation record. Phase 1 does not add shore IDs, occurrence IDs, frozen causal factors, component CFFs, or a fallback global surface rewrite. Its final 3D locality criterion is that the existing orientation-local MUV route acts on the complete parent CFF, with the soft hard-dual changing only the deformation and Laurent endpoint. The per-orientation profiles are the production acceptance test for that rule.

Likewise, the one-second `sysinfo` telemetry paragraph belongs to the explicitly superseded fixed-orientation investigation. It is not a Phase-1 acceptance criterion and no process-monitor dependency or scheduler change is introduced for the soft operator.

The two UV-mass roles are now kept distinct during 3D forest composition. For a fresh active on-shell energy, the ordinary hard chart uses

$$\frac{m_{UV}^2 r^2 + D - m_{UV,exp}^2}{r^2},$$

so the terminal energy contains the actual vacuum mass while the Taylor coefficient retains the expansion-mass provenance. If an active OSE already contains `mUV`, it is an inner-U terminal energy: `mUV` and the active loop are scaled together and its argument is divided by r^2 , without applying the MUV rearrangement again. This recognition is OSE-local and therefore cannot rescale a disconnected sibling. It relies on the current reserved-symbol invariant that `mUV` inside an active OSE argument denotes a terminal vacuum mass; a future nonterminal use there would require explicit provenance.

Finally, a `--per-orientation` UV profile must not construct its summed curve by adding the already exported `f64` orientation values. Local forest terms can cancel by substantially more than machine precision even when every orientation was evaluated with arbitrary precision. The summed curve is therefore evaluated independently with `orientation=None`, which performs the orientation sum before the final downcast. If any individual orientation required the arbitrary-precision retry, the independent summed evaluation is also forced through that precision. Per-orientation diagnostics remain unchanged and retain their own stability metadata.

Single completed 4D counterterm state

This addendum supersedes the historical statements that a selected 4D atom travels with an all-ordinary reference and that both determine its retained expansion coordinates. Those statements remain above as the investigation record. `Local4dCts` and `Full4dCts` now retain one completed atom; each parent applies its selected operator to that atom, and expansion coordinates are derived only from its actual momentum dependence. Only the selected result is finalized and marked.

The `has_soft_ancestry` flag records whether a nonzero S branch has been executed in the recursive contribution. It propagates through later operators and disconnected products. A degree-zero H or a zero S branch does not introduce soft ancestry, and later algebraic cancellation does not erase existing ancestry. This history supports routing and contextual scheme checks without constructing or comparing a second counterterm expression.

The orchestrator owns early validation of unsupported scheme assignments. Direct `uv_limit` calls retain the corresponding inexpensive invariant errors using `has_soft_ancestry`. Ordinary-scheme baselines needed by algebraic and graph comparisons are constructed explicitly in tests. The selected forest operator and its U, S, US, and combined branch provenance are unchanged.